



Chapter # 03

Respiratory System of Man



Student Learning Outcomes

After studying this chapter, students will be able to:

- [B-12-R-01] Define the respiratory surface and list its properties.
- [B-12-R-02] Describe the main structural features and functions of human respiratory system.
- [B-12-R-03] Explain the ventilation mechanism in humans.
- [B-12-R-04] Discuss the transport of oxygen and carbon dioxide through blood.
- [B-12-R-05] Outline the role of respiratory pigments.
- [B-12-R-06] State the causes, symptoms and treatment of upper respiratory tract infections (sinusitis, otitis media) and lower respiratory infections (pneumonia, pulmonary tuberculosis).
- [B-12-R-07] Describe the disorders of lungs (emphysema and COPD)
- [B-12-R-08] List the effects of smoking on respiratory system.

Respiration, like all essential life processes, operates at two distinct levels: the **cellular level** and the **organismic level**. At the **cellular level**, the process is termed **internal respiration**. It is a catabolic reaction involving the enzymatic breakdown of complex organic compounds, such as glucose, into simpler molecules—primarily carbon dioxide and water—accompanied by the release of energy in the form of **ATP** (adenosine triphosphate). This energy fuels various metabolic activities essential for life.

At the **organismic level**, the process is known as **external respiration**, which involves the **exchange of gases** between the organism and its environment. Oxygen is inhaled into the body, transported to tissues, and carbon dioxide—produced as a byproduct of internal respiration—is expelled.

These two levels of respiration are intricately linked. The oxygen taken in during external respiration serves as the final electron acceptor in cellular respiration, while the carbon dioxide generated at the cellular level is removed through the process of breathing.

This chapter explores the **biological, biochemical, and physiological** dimensions of respiration, highlighting the mechanisms, adaptations, and significance of this fundamental life process.

SLO [B-12-R-1]

Define the respiratory surface and list its properties.

PROPERTIES OF RESPIRATORY SURFACE

Definition

The **respiratory surface** is the specific region of an organism where the **actual exchange of gases** with the external environment takes place. This surface plays a central role in facilitating the diffusion of respiratory gases—primarily oxygen and carbon dioxide—between the organism and its surroundings.

Properties of Respiratory Surface

For gas exchange to be efficient, the respiratory surface must exhibit certain critical properties:

- **Moist and Permeable:** Gases dissolve in moisture before diffusing across membranes. A moist surface ensures that oxygen and carbon dioxide can effectively pass through the membrane.
- **Thin Structure:** Diffusion is only efficient over short distances. A thin respiratory membrane (generally less than 1 mm) minimizes the path length for gas molecules, enhancing the rate of exchange.



- **Large Surface Area:** A greater surface area allows for the exchange of a larger volume of gases. This is especially important in active organisms with high metabolic demands.
- **Rich Blood Supply:** A dense capillary network ensures that gases are rapidly transported to and from the respiratory surface, maintaining an effective concentration gradient.
- **Efficient Ventilation Mechanism:** Continuous ventilation replaces oxygen-depleted air or water and removes carbon dioxide. This helps maintain a steep diffusion gradient, which is essential for sustained gas exchange.

The alveoli of human lungs are lined with a surfactant, a film of lipoprotein that lowers the surface tension and prevents them from closing. Surfactant also speeds up the transport of oxygen and carbon dioxide between the air and liquid lining the alveolus and helps to kill any bacteria, which reach the alveoli. Surfactant is constantly being secreted and reabsorbed in a healthy lung.

SLO [B-12-R-2]

Describe the main structural features and functions of human respiratory system.

RESPIRATORY SYSTEM OF MAN

The **respiratory system** is the biological system responsible for the **exchange of gases**—primarily oxygen and carbon dioxide—between the body fluids and the external environment. This system ensures that oxygen reaches the cells for energy production and that carbon dioxide, a metabolic waste product, is efficiently removed.

Anatomically, the human respiratory system is divided into two major regions:

Upper Respiratory Tract

The **upper respiratory tract** includes the **nose**, **nasal cavity**, and **pharynx**. These structures are the first gateway of the respiratory system, performing the essential functions of filtering, warming, and humidifying the air before it passes into the lungs.

1. Nose and Nasal Cavity

The **nose** is the only externally visible structure of the respiratory system. It is made up of **bones**, **cartilage**, and **adipose tissue**, giving it both strength and flexibility.

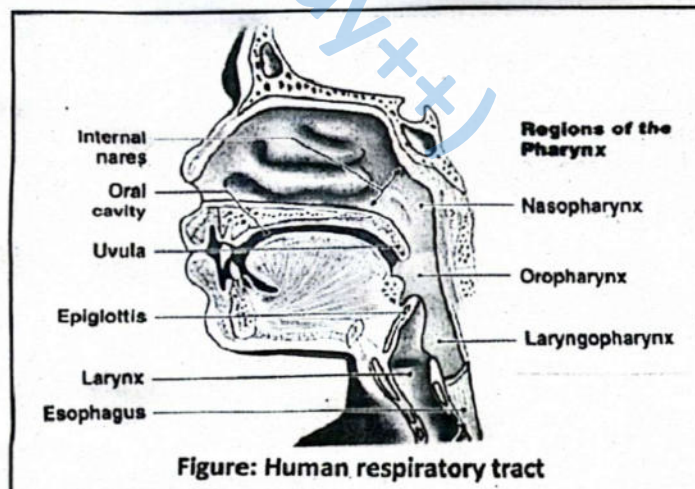
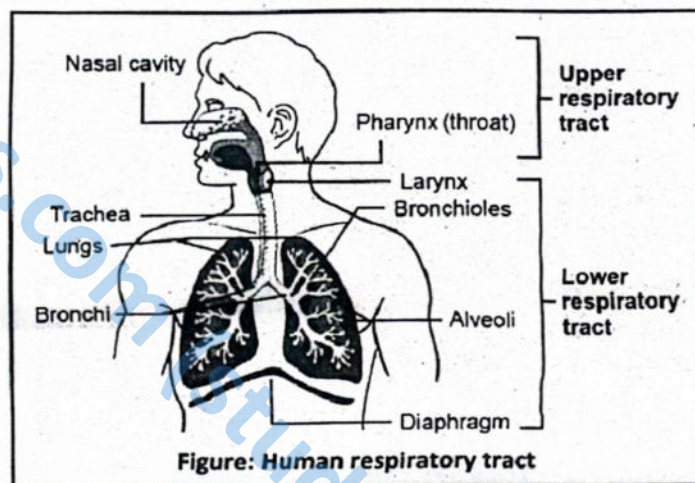
- The two **nostrils (external nares)** open into the **nasal cavity**, which is divided into right and left halves by the **nasal septum**.
- The anterior parts of these cavities, called **vestibules**, are lined with coarse **hairs (vibrissae)** that trap large dust particles.

The inner surfaces of the nasal cavity are lined with a **ciliated mucous membrane**, which plays a crucial **protective and conditioning** role:

► Functions of the Nasal Cavity:

- **Filtration:** Hairs and sticky mucus trap dust, microbes, and airborne pollutants.
- **Ciliary Action:** Cilia continuously sweep mucus and trapped particles backward toward the pharynx, where they are either swallowed or expelled.
- **Humidification:** Mucus adds moisture to incoming air, preventing dryness of delicate lung tissues.

- **Warming of Air:** A rich supply of blood capillaries beneath the mucous membrane warms the inhaled air to about 30°C, close to body temperature.



- **Olfaction:** The upper part of the nasal cavity contains olfactory receptors, allowing the sense of smell, which also helps detect harmful substances.

2. Pharynx

The pharynx is a **funnel-shaped muscular tube** about **12–14 cm** long. It connects the nasal and oral cavities to the **larynx** and **esophagus**, functioning as a common passageway for both **air (respiration)** and **food (digestion)**.

Structurally, the pharynx is divided into three regions:

1. Nasopharynx

- Located behind the nasal cavity.
- Contains openings of the **Eustachian tubes** (auditory tubes), which equalize air pressure in the middle ear.
- Contains the **pharyngeal tonsils (adenoids)**, which provide immune defense against inhaled pathogens.

2. Oropharynx

- Lies behind the oral cavity.
- Serves as a common passageway for both air and food.
- Contains the **palatine tonsils** and **lingual tonsils**, which also have immune functions.

3. Laryngopharynx

- Located behind the larynx and leading into both the **larynx (respiratory system)** and **esophagus (digestive system)**.
- Acts as a crossroad where food is directed toward the esophagus and air toward the larynx.

Lower Respiratory Tract

The **lower respiratory tract** includes the **larynx, trachea, bronchi, bronchioles, alveolar ducts, and alveoli**. Its primary role is to conduct air into the lungs and ensure efficient **gas exchange** between air and blood.

1. Larynx (Voice Box)

The **larynx** lies just below the pharynx and marks the entrance to the trachea. It is a **muscular and cartilaginous chamber** that not only allows air passage but also plays a vital role in **protection and sound production**.

► Functions of the Larynx:

- **Air Conduction:** Provides a passageway for air between the **pharynx and trachea**.
- **Airway Protection:** During swallowing, the **epiglottis** (a flap-like elastic cartilage) folds backward to cover the laryngeal opening, preventing food or liquid from entering the trachea.
- **Voice Production:** The **vocal cords (vocal folds)** inside the larynx vibrate when air passes through them, producing sound. Pitch and tone are controlled by the tension and length of these cords.
- **Voice Modulation:** The **glottis**, which is the opening between the vocal cords, is lined by mucous membrane and adjusts in size to help in voice control and resonance.

2. Trachea (Windpipe)

The **trachea** is a flexible tube about **10–12 cm long** and **2–2.5 cm wide**. It begins at the larynx and extends downward into the thoracic cavity, where it divides into the two primary bronchi.

► Structural Features:

- Made of **15–20 C-shaped cartilage rings** that provide rigidity and prevent collapse.
- The open side of the C-rings faces the esophagus, allowing flexibility during swallowing.
- Lined with **ciliated pseudostratified epithelium** and mucus-secreting goblet cells, which trap dust and microbes.
- **Ciliary action** moves the mucus upward toward the pharynx in what is called the **mucociliary escalator**, keeping the lower tract clean.

Function: Ensures an open and flexible airway, filtering and conducting air to the bronchi.

3. Bronchi and Bronchioles

At the lower end, the trachea bifurcates into the right and left primary bronchi:

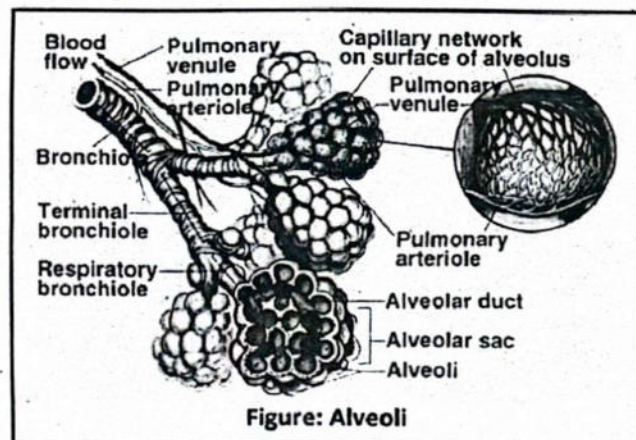
- The **right bronchus** is shorter, wider, and more vertical, making it a common site for inhaled foreign objects.
- The **left bronchus** is longer, narrower, and more horizontal.

► Branching Pattern (Bronchial Tree):

- **Primary (main) bronchi** → one to each lung.
- **Secondary (lobar) bronchi** → 3 in the right lung, 2 in the left, supplying each lobe.
- **Tertiary (segmental) bronchi** → supply bronchopulmonary segments.
- **Bronchioles** → airways less than 1 mm in diameter.
- **Terminal bronchioles** → the last conducting branches before reaching the gas-exchange region.

► Structural Changes along the Tree:

- **Cartilage rings** gradually become **cartilage plates**, then disappear in bronchioles.
- **Smooth muscle** increases in proportion, regulating airflow by constriction (bronchoconstriction) or dilation (bronchodilation).



4. Alveolar Ducts and Alveoli

The terminal bronchioles open into **alveolar ducts**, which lead to grape-like clusters of **alveoli**. These are the microscopic structures where **external respiration** occurs.

► Features of Alveoli:

- Human lungs contain about **700 million alveoli**, providing a vast surface area ($\sim 70 \text{ m}^2$) for gas exchange.
- Each alveolus is lined by a **single layer of squamous epithelial cells** (type I pneumocytes), ensuring minimal diffusion distance ($\sim 0.1 \mu\text{m}$).
- A thin film of **moisture** lines the alveoli, dissolving gases for diffusion.
- Surrounded by a dense network of **capillaries**, enabling close contact between air and blood.
- **Elastic and collagen fibers** in alveolar walls allow alveoli to **expand during inhalation** and **recoil during exhalation**, maintaining breathing efficiency.
- Special **type II pneumocytes** secrete **pulmonary surfactant**, which reduces surface tension and prevents alveolar collapse.

► Functions of Alveoli:

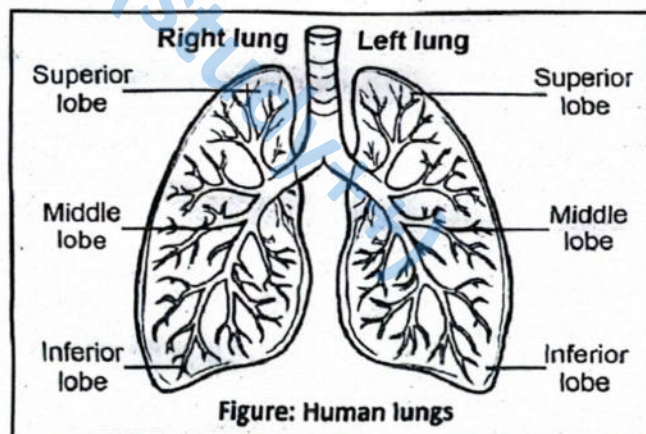
- **Primary sites of gas exchange** (oxygen enters blood, carbon dioxide leaves blood).
- Provide a large, thin, and moist surface adapted for **rapid diffusion of gases**.

External Structure of the Lungs

The **lungs** are the principal organs of respiration. They are **soft, spongy, and elastic** in texture, mainly due to the presence of millions of alveoli and extensive elastic fibers. Their conical shape and light weight allow them to expand and recoil efficiently during breathing.

► Shape and Position

- Each lung is **conical in shape**, designed to fit the thoracic cavity.
- Major features include:
 - **Base** → Broad and concave, resting on the diaphragm to allow smooth expansion during breathing.
 - **Apex** → Tapered upper portion, projecting slightly above the clavicle into the root of the neck.
 - **Costal surface** → Outer, smoothly curved surface that lies in contact with the **ribs and intercostal muscles**.
 - **Mediastinal surface** → Inner surface facing the mediastinum, containing the **hilum**, where bronchi, blood vessels, lymphatics, and nerves enter or leave the lungs.



Lobes of the Lungs

The human lungs are divided into distinct **lobes**, each separated by deep fissures. These lobes ensure efficient distribution of air and blood within the lungs, allowing for optimal gas exchange.

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► Right Lung

- The right lung is **larger in volume** (about 55% of total lung capacity) but **shorter and broader** because the diaphragm rises higher on the right side due to the underlying liver.
- It is divided into **three lobes**:
 1. **Superior lobe** – located at the apex and anterior region.
 2. **Middle lobe** – wedge-shaped, situated between the other two lobes.
 3. **Inferior lobe** – the largest, extending toward the base of the lung.
- These lobes are separated by **two fissures**:
 - **Horizontal fissure** – separates the superior lobe from the middle lobe.
 - **Oblique fissure** – runs diagonally, separating the inferior lobe from both the superior and middle lobes.

► Left Lung

- The left lung is **narrower and slightly longer** because it shares space with the heart within the thoracic cavity.
- It comprises **two lobes**:
 1. **Superior lobe** – forms the upper part and contains the **lingula**, a tongue-like projection that partly corresponds to the middle lobe of the right lung.
 2. **Inferior lobe** – occupies the base.
- The lobes are separated by a single oblique fissure.
- A distinctive feature is the **cardiac notch**, an indentation on the anterior border that accommodates the apex of the heart.

Hilum and Root of the Lung

► Hilum of the Lung

- The **hilum** is a wedge-shaped depression situated on the **mediastinal surface** of each lung.
- It functions as the **gateway** through which all major structures enter or exit the lung, thereby linking the lungs with the rest of the body.
- The hilum is covered by the reflection of the **pleura**, forming a fold known as the **pulmonary ligament**, which provides support and allows movement during respiration.

► Structures Passing Through the Hilum

1. **Bronchi** – main airways that branch from the trachea and carry air into and out of the lungs.
2. **Pulmonary arteries** – deliver **deoxygenated blood** from the right ventricle of the heart to the lungs for oxygenation.
3. **Pulmonary veins** – return **oxygen-rich blood** to the left atrium of the heart.
4. **Lymphatic vessels** – transport lymph and filter harmful substances through lymph nodes, aiding in immune defense.
5. **Nerves** – components of the autonomic nervous system that regulate bronchial muscle tone and vascular diameter.

► Root of the Lung

- The **root of the lung** refers collectively to the bronchi, pulmonary arteries, pulmonary veins, lymphatic vessels, and nerves that pass through the hilum.
- It secures the lung to the **mediastinum** while still allowing flexibility for breathing movements.
- The root is enclosed in a tubular fold of pleura known as the **pleural sleeve**, from which the **pulmonary ligament** descends, helping stabilize the lung in the thoracic cavity.

► Arrangement of Structures

- The arrangement of **bronchi** and vessels in the root differs between the right and left lung:
 - **Right lung** – the bronchus lies **above** the pulmonary artery (hence the mnemonic "**RALS**": **R**ight lung – **A**rtery is **A**nterior, **L**eft lung – **A**rtery is **S**uperior).
 - **Left lung** – the pulmonary artery lies **above** the bronchus.

This anatomical difference is important for surgeons and radiologists.

Pleural Covering

The lungs are delicate and spongy organs, and to protect them, they are enclosed within a specialized double-layered membrane known as the **pleura**. This pleural covering plays an essential role in both protection and the mechanics of breathing.

1. Layers of the Pleura

- **Visceral pleura:** This inner layer is firmly attached to the external surface of the lung, covering every lobe and fissure. It moves with the lung tissue during all respiratory movements.
- **Parietal pleura:** This outer layer lines the thoracic cavity, the upper surface of the diaphragm, and the sides of the mediastinum. It acts as a boundary between the lungs and the walls of the chest cavity.

2. Pleural Cavity

- The space between the visceral and parietal layers is called the **pleural cavity**.
- It is not an empty space but a potential space containing a very thin film of **pleural fluid** secreted by the pleura.

3. Functions of Pleural Fluid

- **Lubrication:** The pleural fluid minimizes friction between the two pleural layers during inhalation and exhalation, allowing smooth and pain-free movement of the lungs.
- **Surface tension:** The fluid creates a slight adhesive force that helps the lungs remain in close contact with the thoracic wall. This ensures that when the chest cavity expands during inspiration, the lungs expand along with it.

4. Clinical Importance

- Disturbances of the pleural cavity can lead to respiratory problems. For example:
 - **Pleurisy:** Inflammation of the pleura, causing painful breathing.
 - **Pleural effusion:** Excess accumulation of fluid in the pleural cavity, restricting lung expansion.
 - **Pneumothorax:** Entry of air into the pleural cavity, which can cause lung collapse.

Comparison of Structure and Functions of Different parts of Respiratory System of Human

Part	Structural Features	Functions
Nasal Cavity	Lined with mucous membrane and ciliated epithelium; contains nasal conchae and hairs	Warms, humidifies, and filters incoming air; traps dust and pathogens via mucus and cilia
Pharynx	Muscular tube divided into nasopharynx, oropharynx, and laryngopharynx	Serves as a common pathway for air and food; involved in vocal resonance
Larynx (Voice Box)	Cartilaginous structure containing vocal cords; connects pharynx to trachea	Produces sound; acts as a passageway for air; prevents food entry into the trachea (epiglottis)
Trachea (Windpipe)	Flexible tube supported by C-shaped cartilage rings; lined with ciliated epithelium	Conducts air to bronchi; cilia and mucus trap and expel foreign particles
Bronchi	Two main branches from trachea, each entering a lung and further dividing into bronchioles	Direct air into lungs; continue filtering air through mucosal lining
Bronchioles	Small branches of bronchi lacking cartilage; have smooth muscle walls	Regulate airflow to alveoli through bronchodilation and bronchoconstriction
Alveoli	Thin-walled, balloon-like sacs surrounded by capillaries	Primary site of gaseous exchange between air and blood (oxygen and carbon dioxide diffusion)
Lungs	Pair of spongy organs enclosed in pleural membranes; contain bronchi, bronchioles, alveoli	Perform pulmonary ventilation and gas exchange; maintain acid-base balance
Diaphragm (accessory)	Dome-shaped skeletal muscle located below the lungs	Facilitates breathing by altering thoracic cavity volume during inspiration and expiration

RESPIRATORY SYSTEM

- Which of the following structures prevents food from entering the trachea during swallowing?
A. Glottis B. Nasopharynx
C. Epiglottis D. Vocal cords
- Which part of the respiratory tract is lined with ciliated mucous membrane that filters, moistens, and warms the air?
A. Larynx B. Nasal cavity
C. Alveoli D. Trachea
- What is the function of surfactant in the alveoli?
A. Increases blood flow
B. Enhances oxygen production
C. Reduces surface tension
D. Regulates temperature
- Which of the following structures lacks cartilage support?
A. Primary bronchi B. Secondary bronchi
C. Terminal bronchioles D. Trachea
- Which region of the pharynx connects directly to the larynx and esophagus?
A. Nasopharynx B. Oropharynx
C. Laryngopharynx D. Tracheopharynx
- How many secondary bronchi are present in the human lungs?
A. 4 (2 per lung) B. 5 (3 right, 2 left)
C. 6 (3 per lung) D. 7 (4 right, 3 left)
- Which structure of the respiratory system contains the vocal cords?
A. Pharynx B. Glottis
C. Larynx D. Trachea
- The walls of alveoli are composed of which type of epithelium?
A. Columnar epithelium B. Cuboidal epithelium
C. Squamous epithelium D. Transitional epithelium
- Which part of the respiratory system contains C-shaped cartilage rings?
A. Larynx B. Bronchioles
C. Trachea D. Alveoli
- The hilum of the lung is best described as:
A. A cartilage-covered ridge
B. A depression for muscle attachment
C. The region where vessels and bronchi enter the lung
D. The alveolar sac's protective coating

Answer Key

1. C 2. B 3. C 4. C 5. C 6. B 7. C 8. C 9. C 10. C

Short Questions

Q.1 What is the role of the nasal cavity in respiration?

Ans. The nasal cavity plays a crucial role in preparing inhaled air before it reaches the delicate lower respiratory passages. It is lined with hair that traps large dust particles and other impurities. The mucous membrane secretes mucus, which moistens the incoming air and captures microbes, pollen, and finer dust particles. Cilia present in the nasal lining beat rhythmically to move trapped debris towards the pharynx, where it can be swallowed or expelled. Additionally, the rich supply of capillaries beneath the mucosa warms the air to nearly body temperature, usually around 30°C, preventing cold air from irritating deeper respiratory tissues. Thus, the nasal cavity filters, moistens, and warms inspired air, ensuring efficient and safe respiration.

MARKING SCHEME (3 Marks)

- Filtering function – 1 mark (0.25 each for hair, mucus, cilia)
- Moistening role – 0.50 mark
- Warming role – 0.50 mark
- Integration/importance in respiration – 1 mark

Q.2 Describe the function of the epiglottis during swallowing.

Ans. The epiglottis is a flap-like structure made of elastic cartilage located just above the glottis in the larynx. Its main function is to safeguard the airway during swallowing. When a person is breathing normally, the epiglottis remains upright, allowing air to pass freely into the trachea. During swallowing, however, the larynx moves upward and the epiglottis bends downward like a lid, covering the glottis. This prevents food particles and liquid from entering the trachea and directs them instead into the esophagus. Once swallowing is complete, the epiglottis returns to its original position, reopening the airway. This mechanism prevents choking, ensures proper passage of food, and maintains separation of the respiratory and digestive tracts.

MARKING SCHEME (3 Marks)

- Location of epiglottis – 1 mark
- Step 1: Upright during breathing – 0.50 mark
- Step 2: Downward bend during swallowing – 0.75 mark
- Step 3: Directs food into esophagus, prevents choking – 0.75 mark

Q.3 Why is the surface area of alveoli important?

Ans. The alveoli are tiny balloon-like sacs at the ends of bronchioles, numbering around **700 million** in the human lungs. Their greatest significance lies in the vast surface area they collectively provide, estimated to be about **70–100 square meters**, roughly the size of a tennis court. This enormous surface area facilitates rapid and efficient gas exchange between air and blood. Oxygen diffuses from alveolar air into the capillary blood, while carbon dioxide diffuses out of blood into the alveoli. If the alveolar surface area were smaller, gas exchange would be slower, reducing oxygen delivery to tissues. Diseases like emphysema reduce alveolar surface area and lead to breathing difficulties, underscoring its critical importance in respiration.

MARKING SCHEME (3 Marks)

- Structure of alveoli (tiny sacs, large number) – 1 mark
- Vast surface area (measurement / significance) – 1 mark
- Role in efficient gas exchange – 1 mark

Q.4 What is surfactant, and what role does it play in the lungs?

Ans. Pulmonary surfactant is a **lipoprotein secretion** produced by specialized alveolar cells (type II pneumocytes) in the lungs. It forms a thin film that coats the inner surface of alveoli. Its primary role is to reduce **surface tension** of the alveolar walls, preventing them from collapsing during exhalation. By doing so, surfactant ensures that alveoli remain open, allowing smooth and continuous exchange of gases with every breath. In addition to its mechanical function, surfactant also plays a defensive role, as it contains proteins with **antimicrobial properties** that protect against lung infections. Lack of surfactant, especially in premature infants, can result in neonatal respiratory distress syndrome, highlighting its vital role in breathing.

MARKING SCHEME (3 Marks)

- Definition (lipoprotein film) – 1 mark
- Role in reducing surface tension, preventing collapse – 1.25 marks
- Role in smooth gas exchange and defense – 0.75 mark

Q.5 Differentiate between primary, secondary, and tertiary bronchi.

Ans. The bronchial tree is a series of branching tubes that distribute air throughout the lungs. The **primary bronchi** are the first branches of the trachea; there are two, one entering each lung. The **secondary bronchi** (also called lobar bronchi) branch from the primary bronchi. They deliver air into each lobe of the lungs: three secondary bronchi serve the three lobes of the right lung, while two serve the two lobes of the left lung. The **tertiary bronchi** (segmental bronchi) arise from secondary bronchi and further divide to ventilate distinct bronchopulmonary segments of each lobe. This branching system ensures efficient distribution of air throughout all lung regions, supporting uniform gas exchange.

MARKING SCHEME (3 Marks)

- Primary bronchi description – 1 mark
- Secondary bronchi description – 1 mark
- Tertiary bronchi description – 1 mark

Q.6 What structural changes occur in the airways as they branch?

Ans. As the respiratory tract branches from bronchi into bronchioles, distinct structural changes take place to support their functions. In the larger bronchi, the airway walls contain C-shaped cartilage rings which gradually change into smaller irregular cartilage plates. By the time the airways reach the bronchioles, cartilage completely disappears, leaving the walls supported mainly by smooth muscle. This smooth muscle plays a regulatory role, allowing the bronchioles to constrict or dilate and thus control airflow into the alveoli. Additionally, the airway diameter progressively decreases with each branching, leading to terminal bronchioles and eventually the alveolar ducts. These changes make the airways lighter, more flexible, and efficient for distributing air to the gas exchange surfaces of the lungs.

MARKING SCHEME (3 Marks)

- Cartilage rings → plates → absent – 1 mark
- Presence of smooth muscle – 1 mark
- Decreasing airway diameter → terminal bronchioles – 1 mark

Q.7 What is the function of the pharynx in respiration?

Ans. The pharynx is a muscular tube located behind the nasal and oral cavities, extending down to the larynx and esophagus. It serves as a shared pathway for both air and food, functioning in both the respiratory and digestive systems. In respiration, it allows inhaled air from the nasal cavity or oral cavity to pass into the larynx and further down into the trachea. The pharynx is divided into three distinct regions: the **nasopharynx**, which connects with the nasal cavity; the **oropharynx**, which lies behind the oral cavity; and the **laryngopharynx**, which opens into the larynx. This arrangement ensures that air is properly directed while swallowing reflexes keep food from entering the respiratory tract.

MARKING SCHEME (3 Marks)

- Location of pharynx – 1 mark
- Respiratory role (passage for air) – 1 mark
- Naming of three regions – 1 mark



Q.8 Describe the structure of the trachea.

Ans. The trachea, commonly called the windpipe, is a tubular structure about 10–12 cm long and 2 cm in diameter, extending from the larynx to the primary bronchi. Its wall is reinforced by 15–20 C-shaped cartilage rings that provide rigidity and prevent collapse during inspiration and expiration. The open ends of these rings face posteriorly and are bridged by smooth muscle (trachealis muscle), which allows flexibility and slight narrowing of the trachea during coughing. The inner lining of the trachea consists of ciliated pseudostratified epithelium with goblet cells that secrete mucus to trap dust and microbes. The coordinated beating of cilia moves trapped particles upward toward the pharynx for expulsion, ensuring clean air reaches the lungs.

MARKING SCHEME (3 Marks)

- Cartilage rings (support) – 1 mark
- Smooth muscle (flexibility, contraction) – 1 mark
- Mucous membrane with cilia (filtering role) – 1 mark

Q.9 What makes the alveolar wall suitable for gas exchange?

Ans. The alveolar wall is highly specialized to allow efficient diffusion of gases. It is composed of a single layer of thin squamous epithelial cells (Type I pneumocytes), making the wall extremely thin—only about 0.1 µm thick. This greatly reduces the diffusion distance for gases. The alveolar surface is kept moist, which helps dissolve gases and facilitates their movement across the membrane. Surrounding each alveolus is a dense capillary network, ensuring a continuous supply of deoxygenated blood and rapid oxygen uptake. Additionally, elastic fibers in the alveolar walls allow stretching during inhalation and recoil during exhalation, maintaining proper airflow. Together, these features make alveoli highly efficient sites for oxygen and carbon dioxide exchange.

MARKING SCHEME (3 Marks)

- Structural adaptation: extreme thinness – 1 mark
- Structural adaptation: moist lining + capillary network – 1 mark
- Functional adaptation: efficient diffusion of gases – 1 mark

Q.10 Explain the anatomical difference between the right and left lungs.

Ans. The two lungs differ structurally due to their position in the thoracic cavity and the presence of the heart. The **right lung** is broader and shorter but has a larger volume. It is divided into **three lobes**—superior, middle, and inferior—separated by horizontal and oblique fissures. The **left lung** is slightly smaller and narrower, as it shares space with the heart, forming a cardiac notch on its medial surface. It has only **two lobes**—superior and inferior—separated by an oblique fissure. Despite these differences, both lungs are spongy in texture and contain a hilum, the entry and exit point for bronchi, blood vessels, lymphatics, and nerves, ensuring coordinated respiratory function.

MARKING SCHEME (3 Marks)

- Right lung anatomy: 3 lobes, larger, fissures – 1.50 marks
- Left lung anatomy: 2 lobes, cardiac notch, smaller – 1.50 marks

SLO [B-12-R-3]

Explain the ventilation mechanism in humans.

VENTILATION MECHANISM IN HUMANS

Ventilation (also called *breathing*) is the mechanical process by which air is moved into and out of the lungs, ensuring a continuous supply of oxygen and removal of carbon dioxide. It occurs in two phases: **inhalation** (inspiration) and **exhalation** (expiration).

1. Inhalation (Inspiration)

Inhalation is the process by which air enters the lungs. It is considered an **active phase of breathing**, because it requires the contraction of respiratory muscles and the expenditure of energy. This process ensures that oxygen-rich air is drawn into the alveoli, where gas exchange takes place.

► Mechanism of Inhalation

1. Contraction of the Diaphragm

- The diaphragm is a dome-shaped sheet of muscle separating the thoracic cavity from the abdominal cavity.
- During inhalation, it **contracts and flattens downward**, increasing the vertical dimension of the thoracic cavity.
- This movement also pushes the abdominal contents slightly downward and forward.



2. Contraction of External Intercostal Muscles

- The external intercostal muscles are located between the ribs.
- When they contract, the ribs are pulled **upward and outward**, which increases the **anteroposterior** (front-to-back) and **transverse** (side-to-side) diameters of the chest cavity.
- This expansion further enlarges the thoracic cavity.

3. Increase in Thoracic Volume

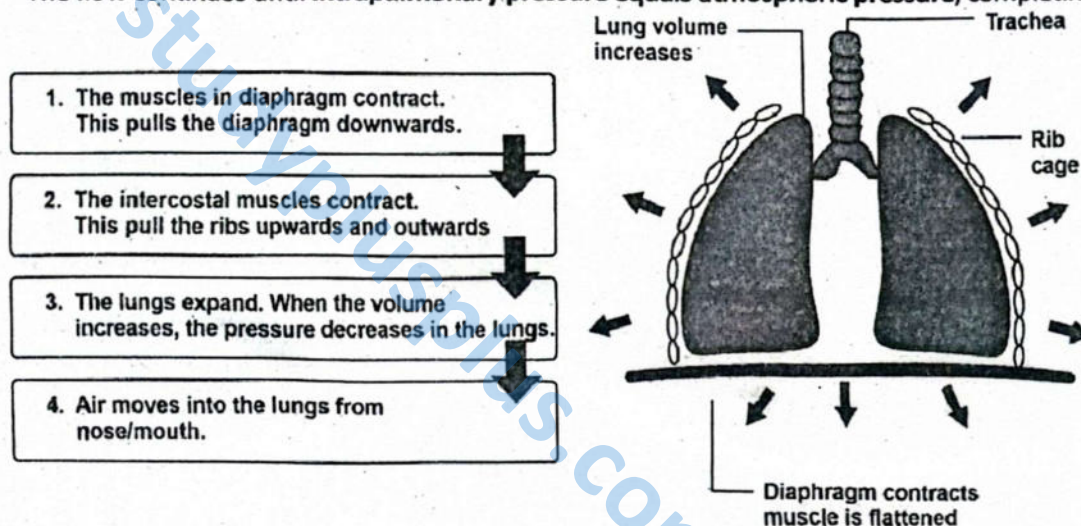
- Combined action of the diaphragm and intercostal muscles results in a significant increase in **thoracic volume** in all dimensions.

4. Decrease in Intrapulmonary Pressure

- According to **Boyle's Law** (which states that pressure and volume are inversely related), an increase in the thoracic volume causes a **drop in intrapulmonary pressure**.
- The pressure inside the lungs becomes slightly **lower than atmospheric pressure**.

5. Entry of Air into the Lungs

- Due to the created pressure gradient, air rushes into the lungs from the atmosphere.
- The flow continues until **intrapulmonary pressure equals atmospheric pressure**, completing inspiration.



2. Exhalation (Expiration)

Exhalation is the process by which air is expelled from the lungs into the atmosphere. Unlike inhalation, it is usually a **passive process** during quiet breathing, because it does not require active muscular effort. Instead, it mainly relies on the natural **elastic recoil** of the lungs and relaxation of the respiratory muscles.

► Mechanism of Normal (Quiet) Exhalation

1. Relaxation of the Diaphragm

- After contraction during inhalation, the diaphragm relaxes and moves back into its **dome-shaped resting position**.
- This reduces the vertical dimension of the thoracic cavity.

2. Relaxation of External Intercostal Muscles

- The external intercostals also relax, causing the ribs to move **downward and inward**.
- As a result, both the anteroposterior and transverse diameters of the thoracic cavity decrease.

3. Decrease in Thoracic Volume

- Relaxation of these muscles leads to a **smaller thoracic cavity**, reducing lung volume.

4. Increase in Intrapulmonary Pressure

- According to **Boyle's Law**, a decrease in volume leads to an **increase in pressure**.
- Intrapulmonary pressure becomes slightly **higher than atmospheric pressure**.

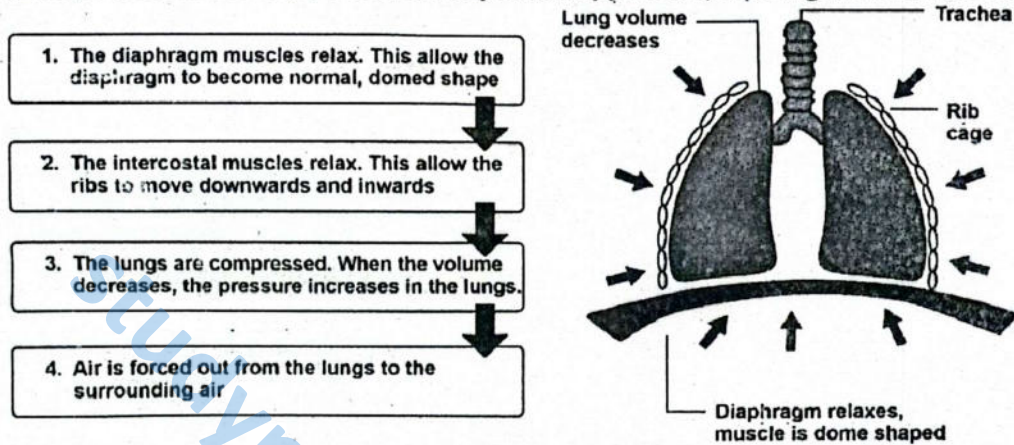
5. Air Flows Out

- Air is forced out of the lungs **down the pressure gradient**, moving from the area of higher pressure (lungs) to lower pressure (atmosphere).

► Forced (Active) Expiration

During activities such as **vigorous exercise, coughing, sneezing, or playing wind instruments**, expiration becomes an **active process**, involving additional muscular effort.

- **Internal intercostal muscles:** Contract to pull the ribs downward and inward more forcefully, reducing thoracic volume.
- **Abdominal muscles:** Contract to push the abdominal contents upward against the diaphragm, causing it to rise further into the thoracic cavity.
- Together, these actions sharply **increase intrapulmonary pressure**, expelling air more forcefully and rapidly.



3. Pressure Relationships in Ventilation

Breathing (ventilation) depends upon the creation of **pressure gradients** between the atmosphere, alveoli, and pleural cavity. Air moves into or out of the lungs by **bulk flow**, always following the rule:

"Air flows from a region of higher pressure to a region of lower pressure."

To understand this, three major pressures are important:

1. Atmospheric Pressure (P_{atm})

- This is the pressure exerted by the air surrounding the body.
- At sea level, it is approximately **760 mmHg** (millimeters of mercury).
- Atmospheric pressure is considered the **reference point** against which other pressures are measured.

2. Intrapulmonary (Alveolar) Pressure (P_{alv})

- This is the pressure inside the **alveoli of the lungs**.
- It changes continuously during breathing:
 - **During inhalation:** P_{alv} falls slightly below P_{atm} (about -1 mmHg), drawing air into the lungs.
 - **During exhalation:** P_{alv} rises slightly above P_{atm} (about $+1$ mmHg), pushing air out of the lungs.
- At the end of each breath, when there is no airflow, P_{alv} becomes equal to P_{atm} .

3. Intrapleural Pressure (P_{ip})

- This is the pressure inside the **pleural cavity**, between the visceral and parietal pleura.
- It is always **slightly negative** compared to both P_{atm} and P_{alv} , usually around **-4 mmHg (≈ 756 mmHg)** at rest.
- The negative pressure is essential because it:
 - Prevents lung collapse.
 - Keeps the lungs closely attached to the thoracic wall, so they expand when the chest expands.

► Importance of Pressure Gradients

- **Inhalation:** Thoracic cavity expands $\rightarrow P_{ip}$ becomes more negative $\rightarrow P_{alv}$ falls below P_{atm} $\rightarrow$ air flows into the lungs.
- **Exhalation:** Thoracic cavity reduces in size $\rightarrow P_{ip}$ becomes less negative $\rightarrow P_{alv}$ rises above P_{atm} $\rightarrow$ air flows out of the lungs.

In summary, ventilation is governed by the relationship among **atmospheric, intrapulmonary, and intrapleural pressures**. These dynamic pressure changes ensure that air moves in and out of the lungs efficiently.

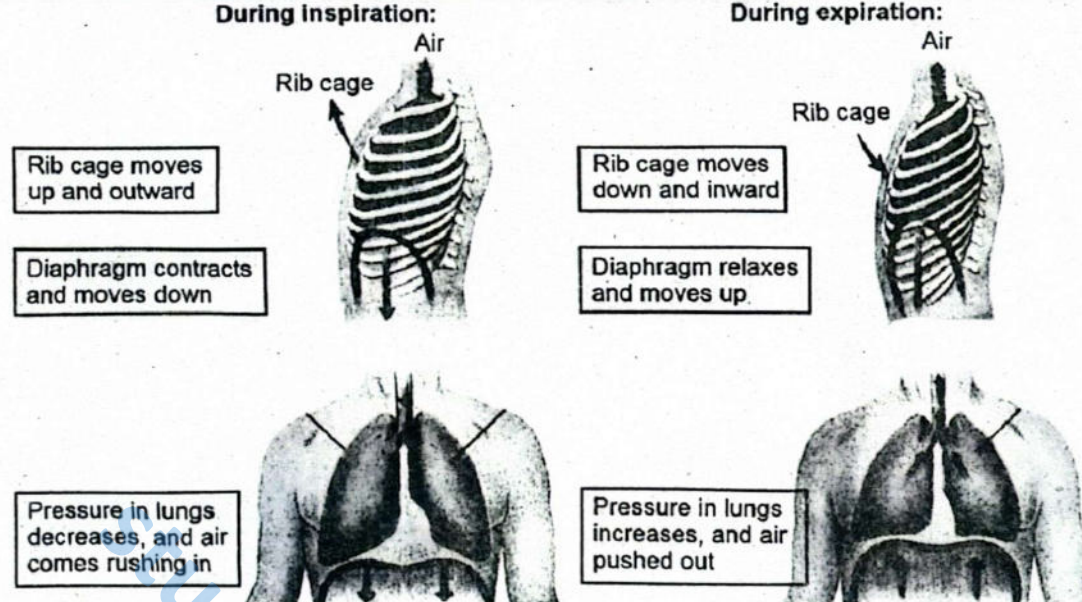


Figure: Mechanism of breathing in human

Feature	Inhalation	Exhalation
Direction of Air	Air enters the lungs	Air leaves the lungs
Diaphragm	Contracts and moves downward	Relaxes and moves upward and dome shaped
Intercostal Muscles	Contract, lifting the ribcage upward and outward	Relax, allowing the ribcage to move downward and inward
Chest Volume	Increases	Decreases
Air Pressure in Lungs	Becomes lower than atmospheric pressure, to bring oxygen into the body	Becomes higher than atmospheric pressure, to remove carbon dioxide from the body
Figure	Lung volume increases Trachea Rib cage Diaphragm contracts muscle is flattened	Lung volume decreases Trachea Rib cage Diaphragm relaxes, muscle is dome shaped

Control of Breathing (Ventilation)

The automatic control of breathing is coordinated by a respiratory centre located in the medulla oblongata of the brainstem. This centre continuously regulates the rate and depth of ventilation to maintain homeostasis.

- The **ventral portion** of the medullary centre functions as the **inspiratory centre**. It sends rhythmic impulses to the diaphragm and intercostal muscles to **initiate and deepen inspiration**.
- The **dorsal and lateral portions** serve as the **expiratory centre**, which **inhibits inspiration and promotes expiration**, particularly during forceful breathing.

Together, these centres adjust ventilation in response to the body's metabolic demands, ensuring that oxygen uptake and carbon dioxide removal are balanced.

► **Voluntary and Higher Brain Control**

While breathing is primarily involuntary, it can be **consciously overridden** via the **cerebral cortex**. This allows individuals to:

- **Alter the rate and depth** of breathing (e.g., during speech, singing, or exercise).
 - **Voluntarily hold their breath**—for example, during swimming or moments of concentration.
- However, breath-holding is **limited by rising carbon dioxide levels** and falling oxygen levels in the blood. If oxygen levels fall too low, unconsciousness occurs, and **automatic breathing resumes** under the control of the respiratory centre.

► **Emotional Influence: The Limbic System**

Breathing patterns can also be modified by **emotional stimuli**, which act through the **limbic system**—a complex network in the brain associated with emotions and behavior. For example:

- **Anxiety** may cause rapid, shallow breathing.
- **Fear or excitement** may accelerate breathing rate.
- **Calmness** can slow breathing.

SLOs Based (MCQs)

1. Which muscles are primarily responsible for increasing the volume of the thoracic cavity during inspiration?
A. Diaphragm and internal intercostals
B. Diaphragm and external intercostals
C. External and internal intercostals
D. Abdominal muscles and diaphragm
2. What happens to the diaphragm during inspiration?
A. It relaxes and becomes more dome-shaped
B. It contracts and flattens
C. It moves upward and reduces lung volume
D. It contracts and forms folds
3. Which of the following occurs during passive expiration?
A. Contraction of diaphragm and external intercostals
B. Relaxation of diaphragm and external intercostals
C. Contraction of all intercostal muscles
D. Active pull by alveoli
4. In which part of the brain is the primary breathing centre located?
A. Cerebellum B. Hypothalamus
C. Medulla oblongata D. Thalamus
5. The inspiratory centre is located in the:
A. Dorsal portion of the medulla
B. Cerebral cortex
C. Ventral portion of the medulla
D. Pons
6. Which muscles contract during forceful expiration?
A. Diaphragm and external intercostals
B. Internal intercostals and abdominal muscles
C. Diaphragm and internal intercostals
D. External intercostals only
7. Which of the following allows voluntary control of breathing?
A. Limbic system B. Cerebral cortex
C. Cerebellum D. Medulla oblongata
8. What happens to intrapulmonary pressure during inspiration?
A. It increases above atmospheric pressure
B. It becomes equal to atmospheric pressure
C. It drops below atmospheric pressure
D. It remains constant
9. How does the limbic system affect breathing?
A. It increases oxygen absorption
B. It influences breathing based on emotional states
C. It initiates reflex coughing
D. It controls muscle contraction in the lungs
10. Which of the following statements is true about expiration during normal breathing?
A. It is an active process
B. It involves contraction of the diaphragm
C. It is a passive process involving muscle relaxation
D. It increases the volume of thoracic cavity

Answer Key

1. B 2. B 3. B 4. C 5. C 6. B 7. B 8. C 9. B 10. C



Q.1 Describe the role of the diaphragm during inspiration.

Ans. The diaphragm is the main muscle of inspiration and plays a crucial role in drawing air into the lungs. During inspiration, the diaphragm **contracts and moves downward**, changing from its dome-shaped resting form to a flattened sheet. This contraction increases the **vertical dimension of the thoracic cavity**, thereby expanding lung volume. As lung volume rises, the **intrapulmonary pressure falls below atmospheric pressure**, creating a negative pressure gradient. Due to this difference, air from the environment rushes into the lungs to equalize pressure. Thus, by its contraction and flattening, the diaphragm ensures the lungs are ventilated efficiently, allowing oxygen to enter and participate in gaseous exchange at the alveoli.

MARKING SCHEME (3 Marks)

- Contraction and flattening – 1 mark
- Thoracic cavity volume increases – 1 mark
- Pressure falls, air enters lungs – 1 mark

Q.2 What changes occur in the thoracic cavity during expiration?

Ans. Expiration is the process of expelling air from the lungs and occurs due to relaxation of respiratory muscles. During quiet breathing, the **diaphragm relaxes and returns to its dome-shaped position**, while the **external intercostal muscles also relax**, allowing the rib cage to descend. This leads to a **decrease in thoracic cavity volume**, which compresses the lungs. As a result, the **intrapulmonary pressure increases above atmospheric pressure**, forcing air out of the lungs. In **forced expiration**, such as during coughing or exercise, the **internal intercostal and abdominal muscles actively contract**, pushing air out more forcefully. These coordinated changes ensure effective removal of carbon dioxide and maintain normal gas exchange.

MARKING SCHEME (3 Marks)

- Diaphragm and external intercostals relax – 1 mark
- Thoracic cavity volume decreases – 1 mark
- Pressure rises, air pushed out – 1 mark

Q.3 What causes air to move into the lungs during inspiration?

Ans. Air enters the lungs because of the development of a **pressure gradient** between the atmosphere and the lungs. When inspiration begins, the **diaphragm contracts and thoracic volume increases**, causing lung expansion. This expansion lowers the **intrapulmonary pressure** below atmospheric pressure. As gases naturally move from a region of higher pressure to lower pressure, air flows down this gradient into the lungs. The greater the increase in thoracic cavity volume, the larger the drop in internal pressure, and the stronger the inflow of air. This mechanism ensures that oxygen-rich air is continuously drawn into the alveoli where diffusion into blood occurs, while maintaining the principle of pressure equalization.

MARKING SCHEME (3 Marks)

- Pressure gradient concept – 1.50 marks
- Thoracic volume increases → pressure drops – 1.50 marks

Q.4 What is the function of the respiratory centre in the medulla oblongata?

Ans. The respiratory centre, located in the **medulla oblongata**, plays a vital role in the automatic regulation of breathing. It ensures that ventilation is continuous and adjusts the **rate and depth of breathing** based on the body's oxygen and carbon dioxide levels. The **ventral respiratory group** is primarily responsible for stimulating inspiration by sending signals to the diaphragm and intercostal muscles. On the other hand, the **dorsal and lateral groups of neurons** help **inhibit inspiration and promote expiration**, especially during passive breathing. Together, these centres coordinate a rhythmic pattern of inhalation and exhalation, ensuring that tissues receive adequate oxygen supply while excess carbon dioxide is eliminated efficiently.

MARKING SCHEME (3 Marks)

- Overall function of centre – 1 mark
- Function of ventral part – 1 mark
- Function of dorsal/lateral part – 1 mark

Q.5 How can breathing be voluntarily controlled?

Ans. Although breathing is largely involuntary, it can be consciously controlled through higher brain centres. The **cerebral cortex** allows voluntary control by overriding the automatic signals from the medulla oblongata. Through this control, a person can consciously regulate breathing for actions such as **speaking, singing, blowing, or holding the breath**. Voluntary control works by **sending direct impulses from the cortex to the respiratory muscles, bypassing automatic pathways**. However, this control is limited: if a person holds their breath too long, the rising carbon dioxide levels and falling oxygen levels in the blood stimulate the medullary respiratory centre, which automatically restarts breathing. This ensures survival by preventing dangerous hypoxia.

MARKING SCHEME (3 Marks)

- Naming the cerebral cortex – 1 mark
- Voluntary control in actions (speaking, holding breath, etc.) – 1 mark
- Automatic resumption when O_2 falls/ CO_2 rises – 1 mark

Q.6 Explain the role of intercostal muscles in ventilation.

Ans. The intercostal muscles, located between the ribs, play an essential role in ventilation by aiding the diaphragm. During **inspiration**, the **external intercostal muscles contract**, pulling the ribs upward and outward. This action expands the thoracic cavity laterally and anteroposteriorly, further increasing lung volume and reducing intrapulmonary pressure so that air enters. During quiet expiration, these muscles simply relax. However, during **forced expiration**, the **internal intercostal muscles contract**, pulling the ribs downward and inward. This decreases thoracic cavity volume, raising pressure inside the lungs and driving air out forcefully. In this way, intercostal muscles act as accessory muscles, ensuring efficient air exchange and supporting both quiet and forceful phases of breathing.

MARKING SCHEME (3 Marks)

- Role of external intercostals (lift ribs, aid inspiration) – 1.50 marks
- Role of internal intercostals (pull ribs down, aid expiration) – 1.50 marks

Q.7 What happens when a person holds their breath for too long?

Ans. When a person deliberately holds their breath, normal gas exchange is disrupted. As time passes, **oxygen levels in the blood decrease** while **carbon dioxide accumulates** because it is not being exhaled. The rising concentration of carbon dioxide stimulates chemoreceptors, which send strong signals to the medullary respiratory centre. If the person continues to resist breathing, the oxygen level may fall to a critically low point while carbon dioxide rises to dangerous levels. This imbalance can cause **dizziness, loss of voluntary control, and eventual unconsciousness**. Once unconscious, voluntary control is removed, and the automatic respiratory centre resumes function, restoring breathing and preventing fatal hypoxia.

MARKING SCHEME (3 Marks)

- Carbon dioxide increases, oxygen decreases – 1 mark
- Effect: dizziness/unconsciousness – 1 mark
- Automatic resumption of breathing – 1 mark

Q.8 How does the limbic system influence breathing?

Ans. The limbic system, a part of the brain involved in emotion and memory, also affects **respiratory patterns**. Emotional states strongly influence breathing rate and depth. For example, **fear, anxiety, or stress** can stimulate the sympathetic nervous system, causing **rapid and shallow breathing (hyperventilation)**. On the other hand, feelings of **calmness, relaxation, or meditation** slow the respiratory rate, producing deeper and steadier breaths. Emotions such as excitement or laughter may temporarily alter breathing rhythms as well. This link between emotional processing and respiratory control explains why breathing exercises are commonly used in stress relief and relaxation techniques. Thus, the limbic system serves as an important modulator of breathing in relation to mental and emotional states.

MARKING SCHEME (3 Marks)

- Function of limbic system in respiration – 1 mark
- Example: fear/anxiety → rapid breathing – 1 mark
- Example: calmness/relaxation → slow breathing – 1 mark

Q.9 Why is expiration usually considered a passive process?

Ans. Expiration, in most normal breathing cycles, is regarded as a passive process. This is because it does not usually require active muscular effort. During expiration, the **diaphragm relaxes** and returns to its dome shape, while the **external intercostal muscles also relax**, lowering the rib cage. As a result, the **elastic recoil of the lungs and thoracic wall** compresses the lungs naturally. This recoil increases intrapulmonary pressure, pushing air out without active contraction. Only in forced expiration, such as during exercise or coughing, do the **internal intercostals and abdominal muscles contract actively**. Under resting conditions, however, expiration occurs passively due to relaxation and recoil, conserving energy during breathing.

MARKING SCHEME (3 Marks)

- Relaxation of diaphragm and intercostals – 1 mark
- Elastic recoil of lungs and thoracic wall – 1 mark
- Air expelled without muscular effort – 1 mark

Q.10 What is the difference between the inspiratory and expiratory centres?

Ans. Breathing is controlled by distinct centres in the medulla oblongata. The **inspiratory centre** is located in the **ventral region** of the medulla. It generates rhythmic **nerve impulses** that stimulate the diaphragm and external intercostals, thereby **initiating inspiration and increasing the rate and depth of breathing** when necessary. In contrast, the **expiratory centre** is situated in the **dorsal and lateral regions** of the medulla. It primarily functions during forced breathing by inhibiting

MARKING SCHEME (3 Marks)

- Role of inspiratory centre (ventral, initiates inspiration) – 1.50 marks
- Role of expiratory centre (dorsal / lateral, supports expiration) – 1.50 marks



inspiratory signals and stimulating internal intercostals and abdominal muscles, thereby **enhancing expiration**. During quiet breathing, this centre remains largely inactive. Together, these two centres coordinate the rhythm of inhalation and exhalation, ensuring efficient and adaptable ventilation.

SLO [B-12-R-4]

Discuss the transport of oxygen and carbon dioxide through blood.

TRANSPORT OF OXYGEN AND CARBON DIOXIDE IN BLOOD

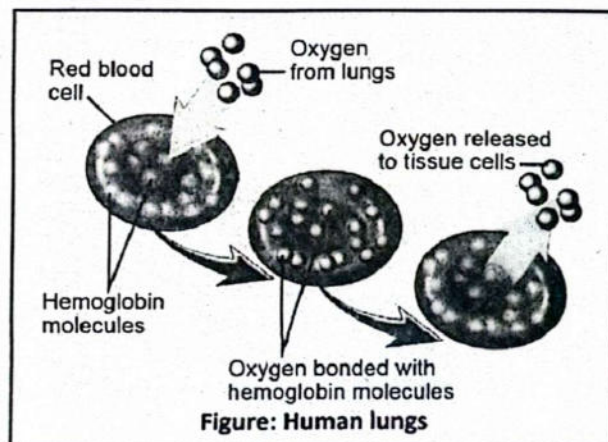
Efficient transport of respiratory gases is essential to maintain cellular respiration and energy production. The blood serves as the medium for carrying **oxygen (O_2)** from the lungs to the body tissues and transporting **carbon dioxide (CO_2)** from the tissues back to the lungs for elimination.

1. Transport of Oxygen

Oxygen is vital for cellular respiration, yet it is only **slightly soluble in plasma**. Therefore, the vast majority of oxygen is carried in the blood bound to **haemoglobin (Hb)** within red blood cells (RBCs). This dual mechanism ensures efficient oxygen delivery from the lungs to tissues with varying demands.

(a) Dissolved Oxygen in Plasma

- A very small portion of oxygen, about **1.5–2%**, is transported as a simple physical solution dissolved in the plasma.
- Though quantitatively minor, this fraction is extremely important because it determines the **partial pressure of oxygen (PO_2)** in the blood.
- PO_2 acts as the driving force for oxygen diffusion:
 - From alveoli into blood (in the lungs).
 - From blood into tissues (in systemic circulation).



(b) Oxyhaemoglobin Formation

- Nearly **98% of oxygen** is carried bound to haemoglobin (Hb) in red blood cells.
- **Structure of haemoglobin:**
 - Each haemoglobin molecule consists of four polypeptide chains.
 - Each chain has a heme group with a central iron atom capable of binding one oxygen molecule.
 - Thus, one haemoglobin molecule can bind up to **four oxygen molecules** to form **oxyhaemoglobin (HbO_2)**.

Reversible Nature of Binding

- The attachment of oxygen to haemoglobin is **reversible**, allowing oxygen to be picked up in the lungs and released in the tissues as required.
- **In the lungs:**
 - High PO_2 and relatively lower CO_2 create favorable conditions for haemoglobin to **bind oxygen efficiently** (oxygen loading).
- **In the tissues:**
 - Low PO_2 , high CO_2 concentration, increased acidity (low pH), and higher temperature favor **release of oxygen** from haemoglobin (oxygen unloading).

This adaptive mechanism ensures oxygen is delivered precisely where it is most needed.

(c) Oxygen–Haemoglobin Dissociation Curve

- The relationship between PO_2 and haemoglobin saturation is shown by the **oxygen–haemoglobin dissociation curve**, which has a characteristic **S-shape (sigmoid curve)**.
- The **sigmoid shape** reflects **cooperative binding**: the binding of one oxygen molecule to haemoglobin increases its affinity for the next, and vice versa.

Factors Influencing Oxygen Release (Shifting the Curve):

1. Partial Pressure of Oxygen (PO_2):

- At high PO_2 (lungs), haemoglobin becomes nearly fully saturated.
- At low PO_2 (tissues), haemoglobin releases oxygen.

2. Carbon Dioxide Levels (PCO_2):

- Increased CO_2 promotes oxygen release — part of the **Bohr effect**.

3. Blood pH:

- A decrease in pH (more acidic conditions due to CO_2 or lactic acid) reduces haemoglobin's affinity for oxygen, enhancing oxygen unloading.

4. Temperature:

- Elevated temperatures, as during muscular activity, also favor oxygen release to tissues.

Physiological Significance

- These shifting conditions ensure that oxygen is held tightly by haemoglobin in the lungs (where it is plentiful) and released efficiently in metabolically active tissues (where it is needed most).

2. Transport of Carbon Dioxide

Carbon dioxide (CO_2) is a waste product of **cellular respiration**, produced continuously in body tissues. Because CO_2 is about **20 times more soluble in plasma than oxygen**, it is transported in the blood more easily. Still, specialized mechanisms are required to carry large amounts efficiently from tissues to the lungs, where it is exhaled.

The three major methods of carbon dioxide transport are:

(a) Dissolved in Plasma (~7–10%)

- A small fraction of CO_2 (about 7–10%) is transported simply dissolved in blood plasma.
- Though this percentage is small, it is important because dissolved CO_2 contributes to the **partial pressure of carbon dioxide (PCO_2)** in the blood, which regulates diffusion between tissues, blood, and alveoli.

(b) Carbaminohaemoglobin (~20–25%)

- About 20–25% of CO_2 binds reversibly with haemoglobin in red blood cells to form **carbaminohaemoglobin ($HbCO_2$)**.
- Unlike oxygen, CO_2 binds to the **globin (protein) part** of haemoglobin, not the heme group.
- **In tissues:**
 - Where PCO_2 is high, more CO_2 binds to haemoglobin, favoring carbaminohaemoglobin formation.
- **In the lungs:**
 - Where PCO_2 is low, CO_2 detaches from haemoglobin and diffuses into the alveoli for exhalation.
- **This process is also influenced by the Haldane effect:**
- Deoxygenated haemoglobin has a greater capacity to bind CO_2 , while oxygenated haemoglobin more readily releases it.
- This ensures efficient CO_2 loading in tissues and unloading in the lungs.

(c) Bicarbonate Ions (~65–70%) — Major Transport Form

- The majority of carbon dioxide (65–70%) is carried as **bicarbonate ions (HCO_3^-)** dissolved in plasma.
- This process occurs primarily inside red blood cells:

1. Formation of Carbonic Acid

- CO_2 diffuses into RBCs and combines with water (H_2O) to form **carbonic acid (H_2CO_3)**.
- This reaction is catalyzed by the enzyme **carbonic anhydrase**, which greatly accelerates the process.

2. Dissociation into Ions

- Carbonic acid quickly dissociates into **hydrogen ions (H^+)** and **bicarbonate ions (HCO_3^-)**.

3. Chloride Shift

- To maintain ionic balance inside the RBC, bicarbonate ions move out into the plasma in exchange for chloride ions (Cl^-).
- This exchange is known as the **chloride shift** or **Hamburger phenomenon**.



- In the lungs, the process reverses:
 - Bicarbonate re-enters RBCs, recombines with H^+ to form CO_2 , which then diffuses into the alveoli and is exhaled.

Transport of Carbon dioxide in Blood

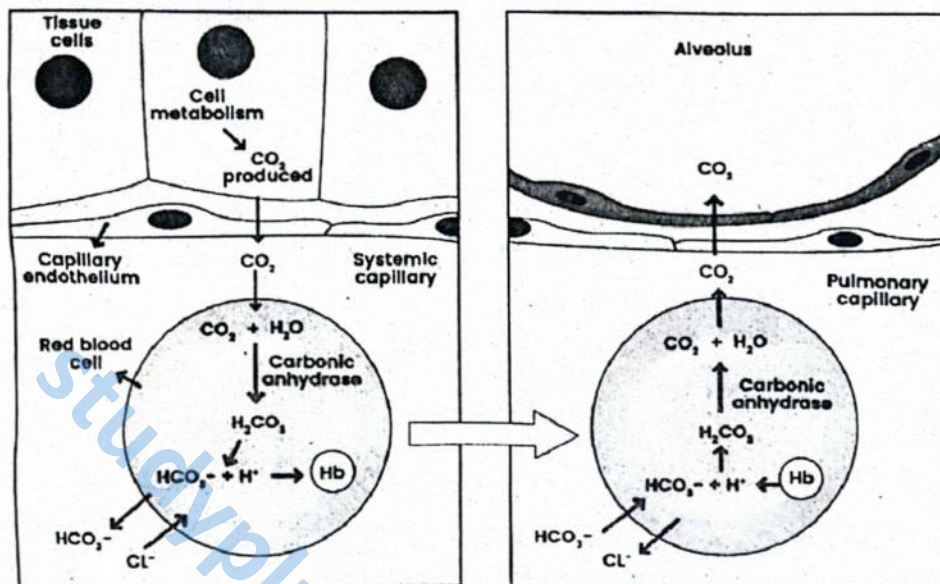


Figure: Transport of CO_2 as bicarbonate ions (a) Transfer of CO_2 from tissues to blood (b) Transfer of CO_2 from blood to lungs.

(I) As Bicarbonate Ions

- During the formation of bicarbonate ions (HCO_3^-) in red blood cells, a compensatory exchange of ions takes place to **maintain electrochemical balance**:
- Chloride ions (Cl^-) enter red blood cells from the plasma via bicarbonate-chloride antiporter proteins. Inside the RBC, Cl^- combines with **potassium ions (K^+)** to form **potassium chloride (KCl)**.
- In the plasma, bicarbonate ions combine with **sodium ions (Na^+)** to form **sodium bicarbonate ($NaHCO_3$)**.
- These interactions help stabilize the **pH of blood around 7.4**, maintained by the **bicarbonate buffer system**.
- Influence of Partial Pressure on CO_2 Transport
- The **movement and transformation of CO_2** are driven by differences in **partial pressure (P_{CO_2})** between tissues, blood, and lungs:
- In **tissues**, PCO_2 is higher than in blood. CO_2 diffuses into RBCs, combines with water to form **bicarbonate ions**, and is transported in this form.
- In the **lungs**, PCO_2 is lower than in the blood. The process reverses:
 - Bicarbonate ions recombine with hydrogen ions (H^+), forming **carbonic acid**, which then *dissociates* into CO_2 and water.
 - CO_2 diffuses into alveoli and is expelled during exhalation.

(II) As Carboxyhaemoglobin

- About **23% of CO_2** is transported by directly binding to the **globin** portion of haemoglobin, forming **carboxyhaemoglobin ($Hb \cdot CO_2$)**.
- In **body tissues**, where PCO_2 is high, CO_2 readily binds with haemoglobin.
- In the **lungs**, where PCO_2 is lower, CO_2 dissociates from haemoglobin and is exhaled.
- This process is **reversible** and regulated by the **relative PCO_2** in different regions of the circulatory system.

(III) Transport as Dissolved CO_2 in Plasma

- A **small fraction—around 7% of total carbon dioxide**—is carried **dissolved directly in blood plasma**.
- Although **not the primary route**, this form contributes to the total CO_2 load and **allows for rapid diffusion** at the gas exchange surfaces.
- Due to its **limited solubility**, this route is relatively **inefficient**, but still functionally significant in CO_2 transport dynamics.

TRANSPORT OF OXYGEN AND CARBON DIOXIDE

- What percentage of oxygen is transported in blood as oxyhaemoglobin?
A. 3% B. 30%
C. 97% D. 100%
- Which enzyme facilitates the reversible binding of oxygen to haemoglobin?
A. Pepsin B. Catalase
C. Carbonic anhydrase D. Lipase
- In what form is the majority of carbon dioxide transported in the blood?
A. Carboxyhaemoglobin
B. Dissolved CO_2 in plasma
C. Bicarbonate ions D. As carbonic acid
- What causes oxygen to be released from oxyhaemoglobin at the tissue level?
A. Decrease in CO_2 levels
B. Increase in pH
C. Increase in hydrogen ion concentration
D. Binding of glucose
- What is the chloride shift in red blood cells?
A. Exchange of sodium and potassium
B. Entry of bicarbonate into RBCs
C. Movement of chloride into RBCs in exchange for bicarbonate
D. Active transport of oxygen
- What is the role of haemoglobin in buffering blood pH?
A. It binds to water
B. It binds to hydrogen ions
C. It converts CO_2 to glucose
D. It activates bicarbonate
- Which ion pairs with bicarbonate in the plasma to form sodium bicarbonate?
A. Cl^- B. Ca^{2+}
C. Na^+ D. K^+
- What happens to bicarbonate ions in the lungs?
A. They are stored in the liver
B. They combine with hydrogen ions to release CO_2
C. They bind to oxygen
D. They are excreted through kidneys
- Where does carboxyhaemoglobin primarily form?
A. In the lungs B. In the alveolar fluid
C. In the tissues D. In the heart
- What is the approximate blood pH maintained by the buffering system?
A. 6.5 B. 7.0
C. 7.4 D. 8.0

Answer Key

1. C	2. C	3. C	4. C	5. C	6. B	7. C	8. B	9. C	10. C
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Short Questions

Q.1 Describe how oxygen is transported in the blood.

Ans. Oxygen transport in blood is primarily carried out by haemoglobin present in red blood cells. About 97% of oxygen molecules bind reversibly with haemoglobin to form oxyhaemoglobin. This binding occurs in the lungs where the partial pressure of oxygen (pO_2) is high, favoring attachment. The remaining 3% of oxygen dissolves directly in plasma, though this plays a minor role. As blood circulates to tissues where the partial pressure of oxygen is lower, haemoglobin releases oxygen, ensuring continuous supply. This property of haemoglobin to bind oxygen where it is abundant and release it where it is deficient ensures efficient delivery to active cells. Thus, oxygen transport depends on haemoglobin and the differences in oxygen pressure.

MARKING SCHEME (3 Marks)

- % oxygen with haemoglobin (97%) – 0.50
- % oxygen dissolved in plasma (3%) – 0.50
- Role of high pO_2 in lungs (binding) – 1
- Role of low pO_2 in tissues (release) – 1

Q.2 Describe the role of bicarbonate ions in carbon dioxide transport.

Ans. Carbon dioxide transport in blood relies heavily on bicarbonate ions. About 70% of CO_2 is carried this way. First, CO_2 diffuses from tissues into red blood cells. Inside, it combines with water under the influence of carbonic anhydrase, forming carbonic acid (H_2CO_3). This unstable compound dissociates into hydrogen ions (H^+) and bicarbonate ions (HCO_3^-). The bicarbonate ions diffuse out into plasma, serving as the

MARKING SCHEME (3 Marks)

- Diffusion of CO_2 into RBCs – 1
- Conversion to carbonic acid – 1
- Formation of H^+ and HCO_3^- , bicarbonate into plasma – 1



major form of CO_2 transport. Meanwhile, hydrogen ions are buffered by haemoglobin to maintain pH. In the lungs, the reverse reaction occurs: bicarbonate re-enters RBCs, reforms CO_2 , and is exhaled. This cycle ensures continuous, efficient transport of CO_2 from tissues to lungs.

Q.3 What is the chloride shift and why is it important?

Ans. The **chloride shift** refers to the exchange of bicarbonate ions (HCO_3^-) leaving red blood cells with chloride ions (Cl^-) entering from plasma. This exchange occurs via a special membrane protein called the bicarbonate-chloride carrier. The shift is essential for maintaining **electrochemical neutrality** inside red blood cells; without it, loss of negatively charged bicarbonate would create imbalance. By ensuring electrical stability, the chloride shift allows continuous movement of bicarbonate into plasma, which is the main form in which CO_2 is carried. It also plays a crucial role in sustaining efficient gas exchange and preventing harmful pH disturbances during CO_2 transport.

MARKING SCHEME (3 Marks)

- Definition of chloride shift – 1
- Maintains electrochemical neutrality – 1
- Importance for continuous CO_2 transport – 1

Q.4 How does haemoglobin help in maintaining blood pH?

Ans. Haemoglobin is not only an oxygen carrier but also a **pH buffer** in the blood. When CO_2 enters RBCs, it forms carbonic acid that dissociates into H^+ and HCO_3^- . The free **hydrogen ions (H^+)** could sharply lower blood pH. To prevent this, haemoglobin binds to these ions, forming **haemoglobinic acid (HHb)**, which stabilizes pH near 7.4. Meanwhile, bicarbonate ions move out into plasma, balanced by the **chloride shift** that maintains electrical neutrality. In the lungs, the process reverses: haemoglobin releases H^+ to recombine with bicarbonate, forming CO_2 for exhalation. This buffering role of haemoglobin is essential for preventing acidosis and ensuring proper cellular activity.

MARKING SCHEME (3 Marks)

- H^+ ions buffered by haemoglobin – 1
- Role of bicarbonate ions – 1
- Chloride shift supports balance – 1

Q.5 Describe how carbon dioxide is transported as carboxyhaemoglobin.

Ans. A significant fraction of carbon dioxide, about 23%, is transported by binding to the **globin portion of haemoglobin**. This compound is called **carboxyhaemoglobin** or carbaminohaemoglobin. This binding occurs mainly in tissues where the **partial pressure of CO_2 is high**, which promotes attachment. Unlike oxygen, CO_2 does not compete directly with haem groups but instead binds to amino groups on globin chains. When blood reaches the lungs, the **partial pressure of CO_2 falls**, and the bound CO_2 dissociates from haemoglobin. It then diffuses into the alveoli to be exhaled. This mechanism allows haemoglobin to carry not just oxygen but also a portion of carbon dioxide efficiently.

MARKING SCHEME (3 Marks)

- % of CO_2 transported this way (23%) – 1
- High pCO_2 in tissues promotes binding – 1
- Low pCO_2 in lungs promotes release – 1

SLO [B-12-R-5]

Outline the role of respiratory pigments.

ROLE OF RESPIRATORY PIGMENTS

Respiratory pigments are coloured molecules that play a vital role in the transport of oxygen within organisms. These pigments **bind reversibly with oxygen**, allowing them to pick up oxygen in regions of high concentration (e.g., lungs or gills) and release it where it is needed (e.g., tissues undergoing respiration). A common feature among all respiratory pigments is the presence of a **coloured, non-protein component**—for example, the **haem group** in haemoglobin.

Two of the most studied respiratory pigments in vertebrates are **haemoglobin** and **myoglobin**.

Haemoglobin

Haemoglobin is a large, globular protein found within red blood cells. It is composed of **four polypeptide chains**, each associated with a **haem group** that contains a central **iron (Fe^{2+}) atom**. This iron atom is capable of binding **one molecule of oxygen**, allowing each haemoglobin molecule to carry up to **four oxygen molecules**.

Oxygen binding to haemoglobin is **reversible** and depends on the **partial pressure of oxygen (pO_2)**. In oxygen-rich environments such as the alveoli of the lungs, haemoglobin readily binds oxygen to form **oxyhaemoglobin**. In tissues with low pO_2 , oxygen is released to support cellular respiration.



Myoglobin

Myoglobin is a structurally simpler pigment found in skeletal and cardiac muscles. It consists of a **single polypeptide chain** and a single **haem group**, allowing it to bind **one molecule of oxygen**. Myoglobin has a **higher affinity for oxygen** than haemoglobin and serves as an **intracellular oxygen reservoir**.

This pigment becomes particularly important when **oxygen levels drop below 20 mmHg**—a condition that may occur during intense muscular activity. At such low oxygen pressures, myoglobin releases its stored oxygen, thereby **sustaining aerobic respiration** in muscle cells when haemoglobin-derived oxygen becomes limited. It is also responsible for the **red colour of muscle tissue**.

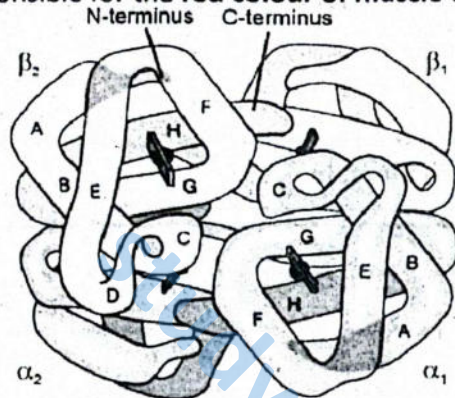


Figure: Haemoglobin

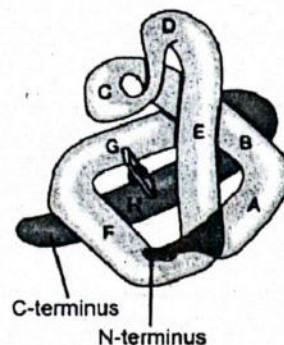


Figure: Myoglobin

Major Roles of Respiratory Pigments

1. Transport of Oxygen

- The primary role is to carry oxygen from the lungs to tissues.
- In humans and other vertebrates, the pigment **haemoglobin (Hb)** is the main oxygen carrier.
- Haemoglobin binds oxygen in the lungs (where PO_2 is high) to form **oxyhaemoglobin** and releases it in the tissues (where PO_2 is low).
- This reversible binding ensures oxygen delivery is adapted to tissue needs.

2. Transport of Carbon Dioxide

- Respiratory pigments also assist in carrying carbon dioxide back from tissues to the lungs.
- Haemoglobin binds with CO_2 to form **carbaminohaemoglobin**, and it also helps buffer hydrogen ions formed during CO_2 transport as bicarbonate.

3. Buffering Role (Maintaining pH)

- Haemoglobin can bind hydrogen ions (H^+), helping to maintain the **acid-base balance** of blood.
- This buffering prevents large fluctuations in blood pH during metabolism.

4. Adaptation to Varying Oxygen Demands

- Respiratory pigments ensure that sufficient oxygen is delivered to tissues even under special conditions, such as:
 - **Exercise** (when tissues require more oxygen).
 - **High altitude** (where atmospheric oxygen is lower).
 - **Diving animals** (which store oxygen in myoglobin for use during long dives).

Feature	Haemoglobin	Myoglobin
Location	Found in red blood cells (erythrocytes).	Found in skeletal and cardiac muscle cells.
Structure	Tetramer (4 polypeptide chains, each with a haem group).	Monomer (single polypeptide chain with one haem group).
Oxygen Binding Capacity	Can bind up to 4 oxygen molecules.	Can bind only 1 oxygen molecule.
Oxygen Affinity	Lower affinity; releases oxygen more readily.	Higher affinity; holds oxygen tightly until very low pO_2 .
Function	Transports oxygen from lungs to tissues.	Stores oxygen in muscles and releases it during low pO_2 .

Oxygen Release Threshold	Releases oxygen at moderate partial pressures of oxygen.	Releases oxygen when pO_2 falls below 20 mmHg.
Role in Respiration	Supports oxygen delivery across the whole body.	Provides oxygen for aerobic respiration in active muscles.
Colour	Gives red colour to blood.	Responsible for red colour of muscle.
Presence of Allosteric Effect	Exhibits cooperative binding (allosteric regulation).	Does not exhibit cooperative binding.
Response to Exercise	Increased oxygen delivery to active tissues.	Acts as an emergency oxygen supply in muscles.

SLO [B-12-R-6]

State the causes, symptoms and treatment of upper respiratory tract infections (sinusitis, otitis media) and lower respiratory infections (pneumonia, pulmonary tuberculosis).

RESPIRATORY TRACT INFECTIONS

Respiratory tract infections are among the **most common human diseases**, affecting millions of people worldwide each year. They may involve either the **upper respiratory tract** (nose, sinuses, pharynx, larynx, and middle ear) or the **lower respiratory tract** (trachea, bronchi, lungs).

1. Upper Respiratory Tract Infections (URTIs)

(a) Sinusitis

Definition:

- Sinusitis is the inflammation or infection of the **paranasal sinuses**, the air-filled cavities surrounding the nasal passages.

Causes:

- Usually occurs after a viral upper respiratory infection ("common cold").
- May be caused by bacteria (e.g., *Streptococcus pneumoniae*, *Haemophilus influenzae*) or, rarely, fungi.
- Predisposing factors: allergies, nasal polyps, deviated septum, or exposure to air pollution/smoke.

Symptoms:

- Facial pain or pressure (especially around eyes, forehead, or cheeks).
- Blocked nose and thick nasal discharge (yellow or green).
- Headache, reduced sense of smell, and sometimes fever.
- Pain worsens when bending forward.

Treatment:

- Most viral sinusitis resolves without antibiotics.
- **Symptomatic relief:** warm steam inhalation, saline sprays, and pain relievers.
- **Bacterial sinusitis:** treated with antibiotics.
- Chronic or recurrent cases may require surgery (functional endoscopic sinus surgery).

(b) Otitis Media

Definition:

- Otitis media is the infection or inflammation of the **middle ear cavity**, located behind the eardrum.

Causes:

- Frequently follows an upper respiratory viral infection, as germs travel from the throat to the middle ear via the Eustachian tube.
- **Common bacteria:** *Streptococcus pneumoniae* and *Haemophilus influenzae*.
- **Risk factors:** bottle-feeding in infants, allergies, exposure to tobacco smoke, and crowded environments (e.g., daycare).

Symptoms:

- Earache (often severe), irritability in children.
- Fever and hearing loss.
- Fluid or pus may leak from the ear if the eardrum ruptures.



Treatment:

- Pain relievers and warm compresses for mild cases.
- Antibiotics are prescribed for bacterial infection.
- Recurrent infections may require insertion of ventilation tubes in the eardrum.

Children are more prone to otitis media due to shorter and more horizontal Eustachian tubes. This is why pediatricians often see ear infections following colds or flu in toddlers.

2. Lower Respiratory Tract Infections (LRTIs)

(a) Pneumonia

Definition:

- Pneumonia is an acute infection of the **alveoli** in the lungs, where air sacs become filled with fluid or pus, making breathing difficult.

Causes:

- Can be caused by bacteria (*Streptococcus pneumoniae* is most common), viruses (influenza, RSV, COVID-19), or fungi (in immunocompromised patients).
- **Risk factors:** smoking, chronic illnesses, old age, weak immunity.

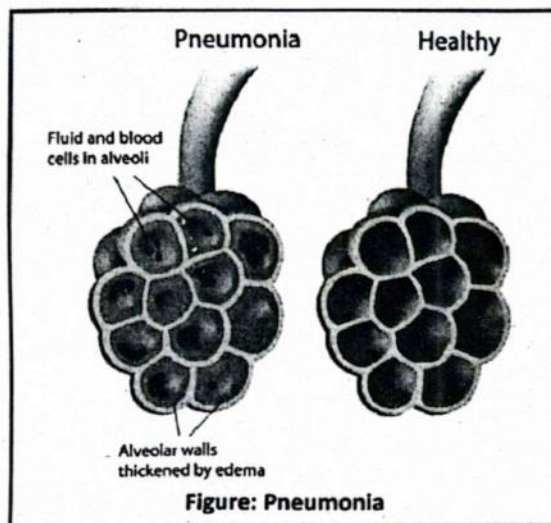
Symptoms:

- High fever, chills, cough with sputum (may be rusty-colored in bacterial cases).
- Chest pain that worsens with breathing or coughing.
- Shortness of breath, rapid breathing, and fatigue.
- In severe cases: bluish lips or nails due to lack of oxygen.

Treatment:

- **Bacterial pneumonia:** treated with antibiotics.
- **Viral pneumonia:** supportive care (fluids, oxygen, antivirals in some cases).
- Hospitalization may be required in severe cases for oxygen therapy.

Pneumonia is a leading cause of hospitalization worldwide, especially in elderly people. Vaccination against pneumococcal bacteria helps prevent it.



(b) Pulmonary Tuberculosis (TB)

Definition:

- Tuberculosis is a **chronic bacterial infection** of the lungs caused by *Mycobacterium tuberculosis*. It primarily affects the lungs but can spread to other organs.

Causes:

- Spread by inhaling airborne droplets released when an infected person coughs, sneezes, or talks.
- **Risk factors:** poor nutrition, overcrowding, HIV/AIDS, and weak immune systems.

Symptoms:

- Persistent cough lasting more than three weeks.
- Coughing up blood-stained sputum.
- Fever, night sweats, unexplained weight loss, fatigue, and chest pain.

Treatment:

- over a long course of **6–9 months**.
- **Directly Observed Treatment, Short-course (DOTS)** is recommended by WHO to ensure patient compliance.

TB remains a **major public health issue in Pakistan and South Asia**. According to WHO, Pakistan is among the high TB burden countries. Awareness, vaccination (BCG vaccine), and early diagnosis are crucial in controlling its spread.

DISORDERS OF THE LUNGS

EMPHYSEMA AND CHRONIC OBSTRUCTIVE PULMONARY DISEASE (COPD)

The lungs are delicate organs responsible for gas exchange. When they are damaged by long-term exposure to irritants such as **cigarette smoke**, **air pollution**, or **occupational dust**, the structure and function of the airways can be permanently altered. Two important chronic respiratory disorders are **emphysema** and **chronic obstructive pulmonary disease (COPD)**. These conditions severely impair breathing and quality of life.

1. Emphysema

Definition:

Emphysema is a chronic lung disease in which the **alveoli** (air sacs) are gradually destroyed, leading to reduced surface area for gas exchange and loss of lung elasticity.

Causes:

- Cigarette smoking (primary cause).
- Long-term exposure to air pollutants, industrial fumes, or dust.
- Genetic deficiency of the enzyme inhibitor **alpha-1 antitrypsin** (rare cause).

Pathophysiology:

- In emphysema, the walls of alveoli break down and merge into larger, irregular air spaces.
- Elastic fibers are destroyed, so the lungs cannot recoil normally.
- This causes air trapping and difficulty in exhaling fully.

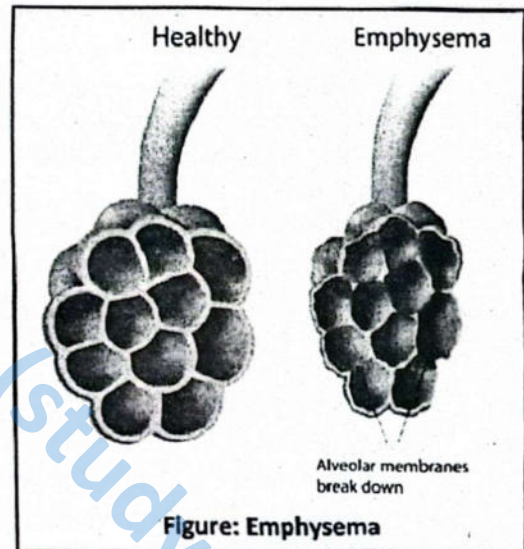
Symptoms:

- Shortness of breath (initially during exertion, later even at rest).
- Chronic cough with little sputum ("dry cough").
- Wheezing and tightness in chest.
- Barrel-shaped chest (due to over-inflated lungs).
- Fatigue and weight loss in advanced stages.

Treatment:

- There is **no cure**, but treatment slows progression.
- Stopping smoking is the most important step.
- Medications: bronchodilators, inhaled steroids, and oxygen therapy.
- Pulmonary rehabilitation exercises to strengthen breathing muscles.
- In severe cases, lung surgery or transplant may be considered.

Emphysema is often called the "smoker's disease." Many long-term smokers develop progressive breathlessness in middle or old age.



2. Chronic Obstructive Pulmonary Disease (COPD)

Definition:

COPD is a **progressive disease** characterized by **long-term obstruction of airflow** in the lungs, making it difficult to breathe. It usually includes **chronic bronchitis** and **emphysema**, often occurring together.

Causes:

- Cigarette smoking is the leading cause (responsible for over 80% of cases).
- Long-term exposure to biomass fuel smoke (common in rural households using wood or coal for cooking).
- Air pollution and occupational dust/fumes.

- Genetic predisposition (e.g., alpha-1 antitrypsin deficiency).

Pathophysiology:

- In **chronic bronchitis**, inflammation and mucus production narrow the airways.
- In **emphysema**, alveoli are destroyed and lose elasticity.
- Together, these changes reduce airflow and impair oxygen exchange.

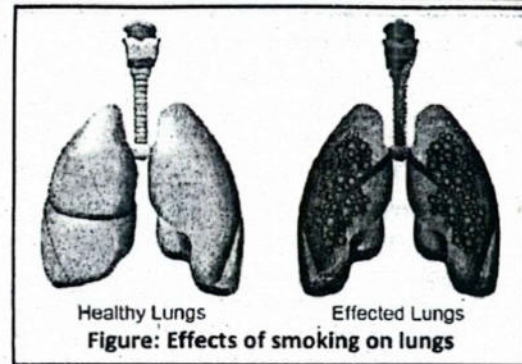
Symptoms:

- Persistent cough with sputum (often called a "smoker's cough").
- Shortness of breath, especially on exertion.
- Wheezing and chest tightness.
- Frequent respiratory infections.
- In severe cases: bluish skin or lips (cyanosis) due to low oxygen levels.

Treatment:

- No complete cure; management aims to reduce symptoms and prevent complications.
- Quitting smoking is essential.
- Bronchodilator inhalers to relax airway muscles.
- Inhaled steroids to reduce inflammation.
- Long-term oxygen therapy for advanced COPD.
- Pulmonary rehabilitation, healthy nutrition, and exercise to improve quality of life.

COPD is among the **leading causes of death worldwide**, particularly in countries with high rates of smoking and pollution exposure. In Pakistan, COPD cases are rising due to **tobacco use and indoor pollution** from cooking fires.



SLO [B-12-R-8]

List the effects of smoking on respiratory system.

EFFECTS OF SMOKING ON THE RESPIRATORY SYSTEM

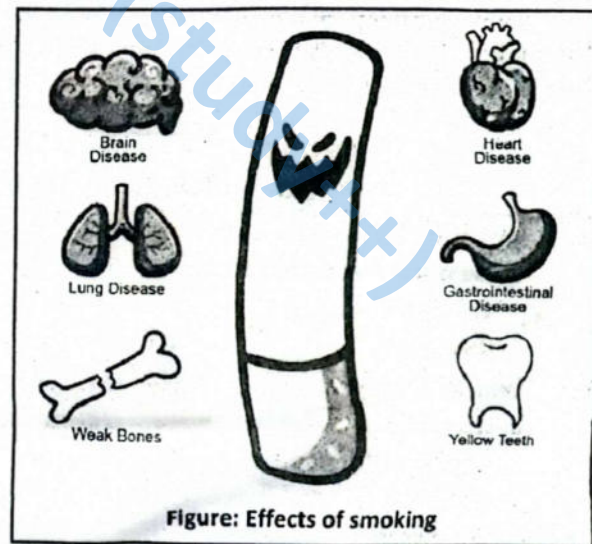
Smoking is one of the most harmful lifestyle habits and a leading cause of respiratory illness worldwide. Tobacco smoke contains thousands of toxic substances such as nicotine, tar, and carbon monoxide, along with many cancer-causing chemicals. These substances damage the air passages, alveoli, and protective mechanisms of the lungs, producing both short-term irritation and long-term chronic diseases.

► Irritation and Inflammation of Airways

When a person inhales cigarette smoke, the chemicals irritate the lining of the trachea and bronchi. This leads to swelling and inflammation of the airways, accompanied by excessive mucus secretion. Over time, this constant irritation causes a persistent cough, often referred to as a "smoker's cough," which is one of the earliest warning signs of lung damage.

► Paralysis of Cilia

The respiratory tract is normally lined with tiny hair-like structures called cilia, which move rhythmically to sweep out dust, microbes, and mucus. Smoking paralyzes or even destroys these cilia. As a result, harmful particles remain trapped in the lungs instead of being cleared out. This greatly increases the risk of respiratory infections such as bronchitis and pneumonia, since the lungs lose their natural defense system.



► **Chronic Bronchitis**

With long-term smoking, the bronchi remain chronically inflamed and produce large amounts of thick mucus. The narrowed and clogged airways make breathing difficult and cause constant coughing. This condition, known as chronic bronchitis, is commonly seen in long-term smokers who experience frequent chest infections and reduced lung efficiency.

► **Emphysema**

One of the most serious effects of smoking is emphysema, in which the walls of the alveoli (air sacs) are progressively destroyed. Instead of many small alveoli providing a large surface area for gas exchange, the alveoli fuse into larger, less efficient sacs. This reduces the surface area for oxygen absorption, leading to severe shortness of breath. Patients with emphysema often develop a "barrel-shaped chest" and may become dependent on supplemental oxygen in advanced stages.

► **Chronic Obstructive Pulmonary Disease (COPD)**

Chronic bronchitis and emphysema together are referred to as Chronic Obstructive Pulmonary Disease (COPD). Smoking is the leading cause of COPD worldwide. It is a progressive disease in which airflow becomes permanently obstructed, making exhalation difficult. Patients with COPD suffer from constant breathlessness, wheezing, and greatly reduced exercise tolerance.

► **Lung Cancer**

Smoking is the major cause of lung cancer, responsible for over 80% of cases globally. Carcinogens present in tobacco smoke damage the DNA of lung cells, leading to uncontrolled cell growth. Lung cancer is particularly dangerous because symptoms often appear late, when the disease has already spread. This makes survival rates very low compared to many other cancers.

► **Reduced Oxygen Transport**

Another harmful effect of smoking is reduced oxygen transport in the blood. Carbon monoxide, a poisonous gas in cigarette smoke, binds to hemoglobin with much greater affinity than oxygen. This prevents hemoglobin from carrying oxygen efficiently, depriving tissues of oxygen. Smokers may experience fatigue, dizziness, and a higher risk of heart and brain complications due to reduced oxygen supply.

► **Increased Susceptibility to Infections**

Since smoking damages both the cilia and the immune defenses of the respiratory system, smokers are far more prone to infections such as tuberculosis, influenza, and pneumonia. Healing of damaged respiratory tissues is also slower in smokers, further worsening recovery from illness.

According to the World Health Organization (WHO), tobacco use kills more than 8 million people every year, with respiratory diseases being a major cause of death. In countries like Pakistan, smoking-related illnesses such as COPD, lung cancer, and tuberculosis place a heavy burden on public health systems. Many of these conditions are preventable if smoking is avoided or stopped early in life.

Smoking affects the respiratory system by irritating and inflaming the airways, paralyzing cilia, causing chronic bronchitis, damaging alveoli in emphysema, leading to COPD, and greatly increasing the risk of lung cancer. It also reduces oxygen transport due to carbon monoxide and makes the body more vulnerable to infections.

SLOs Based (MCQs)

DISORDERS OF RESPIRATORY SYSTEM OF MAN

- | | |
|---|--|
| <p>1. Which of the following is the most common cause of chronic bronchitis and emphysema?</p> <p>A. Viral infections B. Smoking
C. Dust mites D. Cold weather</p> <p>2. Which disorder is characterized by the destruction of alveolar walls, reducing the surface area for gas exchange?</p> <p>A. Pneumonia B. Otitis media
C. Emphysema D. Tuberculosis</p> | <p>3. The Directly Observed Treatment Short Course (DOTS) is used in the treatment of:</p> <p>A. Pneumonia B. Tuberculosis
C. COPD D. Sinusitis</p> <p>4. In pneumonia, which part of the respiratory system primarily gets inflamed?</p> <p>A. Trachea B. Alveoli
C. Larynx D. Sinuses</p> <p>5. Which of the following is a lower respiratory tract infection?</p> |
|---|--|



- A. Sinusitis
C. COPD
6. **Mycobacterium tuberculosis** primarily affects which organ?
- A. Liver
C. Lungs
- B. Otitis media
D. Pharyngitis
- B. Kidney
D. Heart

7. Which of the following is NOT a typical symptom of emphysema?
- A. Shortness of breath
B. Chest tightness
C. Productive cough with blood
D. Fatigue

Answer Key

1. B 2. C 3. B 4. B 5. C 6. C 7. C

Short Questions

Q.1 What is emphysema and what causes it?

Ans. Emphysema is a **chronic lung disease** in which the **alveolar walls are destroyed**, reducing the surface area for gas exchange. As a result, patients experience difficulty in exhaling air fully, leading to breathlessness and low oxygen supply. The main cause of emphysema is **long-term exposure to harmful substances**, especially **cigarette smoke**, which contains toxic chemicals that weaken lung tissue. Other causes include air pollution, industrial dust, and genetic factors such as **alpha-1 antitrypsin deficiency**. The damage to alveoli is **irreversible**, making the disease progressive and disabling. Because the lungs lose elasticity, patients often struggle with chronic coughing and wheezing. Thus, emphysema is a preventable but serious respiratory condition.

MARKING SCHEME (3 Marks)

- Definition of emphysema – 1
- Main cause (smoking / pollution / irritants) – 1
- Effect (reduced gas exchange / breathlessness) – 1

Q.2 How does pulmonary tuberculosis affect the body?

Ans. Pulmonary tuberculosis (TB) is a **lung disease** caused by the bacterium **Mycobacterium tuberculosis**. It spreads through airborne droplets when an infected person coughs or sneezes. The bacteria invade lung tissue, causing the formation of **tubercles (nodules)** and progressive lung damage. Common symptoms include a **persistent cough** (sometimes with **blood-tinged sputum**), **low-grade fever**, **night sweats**, **weight loss**, and **fatigue**. If untreated, TB can spread beyond the lungs to affect other organs, leading to serious complications. Treatment requires long-term use of antibiotics under the **DOTS program** (Directly Observed Treatment, Short-course), which ensures proper drug intake. Without treatment, TB can be fatal, but with proper care it is curable.

MARKING SCHEME (3 Marks)

- Cause (Mycobacterium tuberculosis) – 1
- Symptoms (cough, fever, night sweats, weight loss) – 1
- Effect (lung damage / systemic spread) – 1

Q.3 What are the typical symptoms of pneumonia?

Ans. Pneumonia is an **infection of the lungs** in which the **alveoli become inflamed and filled with fluid or pus**, reducing the efficiency of gas exchange. This results in **difficulty breathing and low oxygen supply** to the body. The typical symptoms include **chest pain**, **fever**, **chills**, **productive cough with mucus**, **shortness of breath**, and **fatigue**. In severe cases, lips and fingertips may turn bluish due to lack of oxygen. Pneumonia can be caused by bacteria (commonly *Streptococcus pneumoniae*), viruses, or fungi. It can affect people of all ages, but infants, elderly individuals, and immunocompromised patients are at higher risk. Early diagnosis and treatment are essential to prevent complications such as pleural effusion or sepsis.

MARKING SCHEME (3 Marks)

- Any six symptoms (0.50 each × 6) – 3

Q.4 Differentiate between COPD and pneumonia.

Ans. COPD (Chronic Obstructive Pulmonary Disease) is a **long-term progressive disease** that includes chronic bronchitis and emphysema. It is mainly caused by **smoking** or prolonged exposure to air pollutants. COPD develops slowly, causing symptoms such as chronic cough, wheezing, and shortness of breath due to irreversible damage to lung tissue.

On the other hand, **pneumonia** is an **acute infection of the alveoli**, often caused by **bacteria, viruses, or fungi**. It leads to inflammation and fluid buildup in alveoli, impairing oxygen exchange. Unlike COPD, pneumonia can often be treated successfully with **antibiotics or antivirals**, and recovery is usually complete. Thus, COPD is a chronic progressive condition, while pneumonia is an acute but often **curable** infection.

MARKING SCHEME (3 Marks)

- Definition and cause of COPD – 1.50
- Definition and cause of pneumonia – 1.50



Q.5 What is the role of the DOTS strategy in TB management?

Ans. DOTS (Directly Observed Treatment, Short-course) is the global strategy recommended by WHO for the effective management of tuberculosis. Under this program, patients are supervised by health workers or trained volunteers to ensure that they take their medicines regularly and complete the course. This prevents patients from stopping treatment prematurely, which is a major cause of drug-resistant TB. DOTS improves cure rates, reduces transmission in the community, and lowers TB-related deaths. It also ensures early diagnosis, proper record keeping, and consistent supply of anti-TB drugs. By focusing on both treatment and prevention of resistance, DOTS has become the cornerstone of TB control worldwide.

MARKING SCHEME (3 Marks)

- Definition of DOTS – 1
- Description of supervised strategy – 1
- Effects (cure, reduced spread, prevent resistance) – 1

Q.6 How does smoking affect the respiratory system?

Ans. Smoking has serious negative effects on the respiratory system. First, the chemicals in cigarette smoke damage the cilia lining the respiratory tract. Since cilia normally clear mucus and trapped particles, their destruction allows harmful substances to accumulate. Second, smoking increases mucus production, leading to persistent coughing and infections. Third, smoke destroys alveolar walls, reducing the surface area for gas exchange, which contributes to diseases such as emphysema. Additionally, smoking causes chronic irritation of the bronchi, resulting in chronic bronchitis. Long-term smoking also greatly increases the risk of lung cancer due to carcinogens present in tobacco. Thus, smoking leads to progressive and often irreversible respiratory diseases, making it a major preventable cause of death.

MARKING SCHEME (3 Marks)

- Cilia destruction – 1
- Increased mucus production – 1
- Damage to alveoli / risk of chronic diseases – 1

Exercise

I

Multiple Choice Questions (MCQs)

❖ Select the correct Answer:

1. When blood leaves the capillary bed most of the carbon dioxide is in the form of
A. carbonate ions B. bicarbonate ions
C. hydrogen ions D. hydroxyl ions
2. When you inhale, the diaphragm
A. relaxes and moves upward
B. relaxes and moves downward
C. contracts and moves upward
D. contracts and moves downward
3. With which other system do specialised respiratory systems most closely interface in exchanging gases between the cells and the environment?
A. the skin B. the excretory system
C. the circulatory system D. the muscular system
4. Which of the following is the respiratory surface in human respiratory system:
A. larynx B. trachea
C. bronchi D. alveoli
5. How is most of the oxygen transported in the blood?
A. dissolved in plasma B. bound to haemoglobin
C. as bicarbonate D. dissolved in water
6. The lateral walls of the chest cavity of man are composed of the:
A. ribs B. intercostal muscles
C. ribs and intercostal muscles
D. ribs, intercostal muscles and diaphragm
7. Which of the following factors is the most effective in accelerating the rate of breathing in man?
A. a lack of oxygen in the blood
B. a lack of oxygen in the tissues
C. an excess of carbon dioxide in the lungs
D. an excess of carbon dioxide in the blood
8. Which of the following changes will increase the body's rate of carbon dioxide excretion into the alveoli?
A. holding the breath
B. the breakdown of alveolar tissue as a result of disease
C. a decrease in the partial pressure of carbon dioxide in the alveolar air
D. a decrease in the pulmonary circulation
9. Breathing is an example of
A. counter current exchange
B. cellular respiration
C. ventilation D. diffusion



10. Which sequence most accurately describes the sequence of airflow in the human respiratory system?
1. pharynx 2. bronchus 3. trachea 4. larynx 5. alveolus 6. bronchiole
A. 4, 1, 3, 2, 5, 6 B. 1, 4, 3, 2, 5, 6
C. 4, 1, 3, 2, 6, 5 D. 1, 4, 3, 2, 6, 5
11. Which one of the following changes takes place during inspiration?
A. Decrease in thoracic cavity
B. Relaxation in diaphragm
C. Relaxation in external intercostal muscles
D. Sternum moves towards ventral and anterior direction
12. The volume of air that can be exhaled after normal exhalation is the
A. tidal volume B. residual volume
C. inspiratory reserve volume
D. expiratory reserve volume
13. Haemoglobin
A. combine reversely with only oxygen
C. attach to the alveolar wall
B. all have four heme group
D. none of them
14. The maximum volume of air contained in the lung by a full forced inhalation is called
A. Tidal volume B. Vital capacity
C. Ventilation rate D. Total lung capacity
15. Dissociation of O_2 from oxyhaemoglobin is facilitated by:
A. decreased temperature B. decreases H^+
C. decreased PO_2 D. exercise
16. Which of these correctly orders the structures through which air passes during inhalation?
A. pharynx → trachea → larynx → bronchi
B. pharynx → larynx → trachea → bronchi
C. larynx → pharynx → bronchi → trachea
D. larynx → pharynx → trachea → bronchi
17. The pharynx is also known as the:
A. windpipe B. trachea
C. voice box D. throat
18. What is the correct path air takes when it enters the trachea on its way to the lungs?
A. bronchi / bronchioles / pulmonary capillaries / alveoli
B. bronchioles / bronchi / alveoli / pulmonary capillaries
C. bronchi / pulmonary capillaries / alveoli / bronchioles
D. bronchi / bronchioles / alveoli / pulmonary capillaries
19. Which of the following is correct for the partial pressure of oxygen in alveoli?
A. less than carbon dioxide
B. less than the blood
C. more than the blood
D. equal to that of the blood
20. Which of the following is the respiratory surface in human respiratory system:
A. larynx B. trachea
C. bronchi D. alveoli

Answer Key

1. B	2. D	3. C	4. D	5. B	6. C	7. D	8. C	9. C	10. D
11. D	12. D	13. B	14. D	15. D	16. B	17. D	18. D	19. C	20. D

II

Short Answer Questions

Q.1 What is respiratory surface? Write the properties of respiratory surface.

Ans. A respiratory surface is the specialized biological interface where the exchange of oxygen and carbon dioxide occurs between the body and the external environment. In humans, the alveoli of the lungs serve as the main respiratory surface, while in other organisms it may be the skin, gills, or tracheal system. For efficient gaseous exchange, the respiratory surface must possess certain properties. It should be thin to allow rapid diffusion of gases, moist so that gases can dissolve before diffusing, and richly supplied with blood capillaries to carry oxygen to tissues and remove carbon dioxide. Moreover, it must have a large surface area to meet the oxygen requirements of the body.

MARKING SCHEME (3 Marks)

- Definition of respiratory surface – 1
- Properties (thin, moist, vascularized, large surface area) – 2

Q.2 What organs constitute the respiratory system?

Ans. The human respiratory system is a complex arrangement of organs that enable breathing and gaseous exchange. The process begins at the nose and nasal cavity, where air is filtered, moistened, and warmed. From there, air passes through the pharynx and into the larynx (voice box). The air then moves into the trachea (windpipe), which divides into right and left bronchi, further branching into smaller bronchioles. At the end of bronchioles are the alveoli, the main sites for gaseous exchange. The lungs house these alveoli and serve as the primary respiratory organs.

MARKING SCHEME (3 Marks)

- Identification of main respiratory organs (nasal cavity, pharynx, larynx, trachea, bronchi, bronchioles, alveoli) – 2
- Supporting organs (lungs, diaphragm, intercostal muscles) – 1



Supporting structures such as the diaphragm and intercostal muscles help in inhalation and exhalation by creating pressure changes in the thoracic cavity.

Q.3 How do nose and nasal cavity function in filtering the incoming air?

Ans. The nose and nasal cavity are the first entry points of air into the respiratory system and perform important protective functions. The air entering through the nostrils is warmed by the blood vessels in the nasal lining, ensuring that cold air does not damage delicate alveoli. At the same time, the mucous membrane of the nasal cavity moistens the air, preventing dryness in the respiratory tract. Most importantly, the mucus traps dust particles, microbes, and pollutants, keeping them from entering deeper parts of the lungs. The cilia (tiny hair-like structures) lining the nasal passage then move these trapped particles toward the throat, where they are either swallowed or expelled. This ensures that only clean air reaches the lungs.

MARKING SCHEME (3 Marks)

- Warming and moistening of air – 1
- Role of mucus in trapping dust/microbes – 1
- Role of cilia in clearing particles – 1

Q.4 What is the role of pharynx in human respiration?

Ans. The pharynx is a muscular tube located behind the nasal and oral cavities. It serves as a common pathway for both air and food. In respiration, the pharynx plays a vital role by conducting inhaled air from the nasal cavity to the larynx, ensuring a continuous passage toward the trachea and lungs. It also helps in proper routing of air, so that inhaled air does not enter the digestive tract. During swallowing, the epiglottis prevents food from entering the airway, keeping the respiratory passage clear. Thus, the pharynx not only functions as a connecting chamber but also ensures safe and efficient air passage in the respiratory process.

MARKING SCHEME (3 Marks)

- Definition/structure of pharynx – 1
- Function in passage of air – 1
- Role in preventing entry of food – 1

Q.5 Describe the function of the human larynx.

Ans. The larynx, commonly known as the voice box, is a cartilaginous structure located between the pharynx and trachea. Its primary function is to produce sound, as it houses the vocal cords, which vibrate when air passes through, enabling speech and vocalization. In addition, the larynx plays an important role in protecting the lower respiratory tract. During swallowing, the epiglottis (a flap of tissue) closes over the laryngeal opening to prevent food or liquid from entering the trachea. The larynx also helps regulate airflow during breathing, ensuring smooth passage of air into the trachea and lungs. Hence, the larynx has a dual role in both voice production and airway protection.

MARKING SCHEME (3 Marks)

- Connection between pharynx and trachea – 0.50
- Role in voice production (vocal cords) – 1
- Role in protecting airway during swallowing – 1.50

Q.6 Describe the structure and function of alveoli.

Ans. The alveoli are microscopic, balloon-like air sacs located at the ends of bronchioles in the lungs. Each human lung contains about 300 million alveoli, providing a vast surface area for gas exchange. Their walls are extremely thin (one cell thick) and kept moist, allowing oxygen and carbon dioxide to dissolve and diffuse efficiently. Surrounding each alveolus is a dense network of capillaries, enabling close contact between air and blood. During gas exchange, oxygen diffuses from the alveolar air into the blood, while carbon dioxide diffuses from the blood into the alveolar air for exhalation. Thus, alveoli form the essential respiratory surface of humans.

MARKING SCHEME (3 Marks)

- Structure: balloon-like sacs, thin and moist walls – 1
- Rich supply of capillaries – 1
- Function in oxygen uptake and CO₂ removal – 1

Q.7 How do the contraction and relaxation of human lungs take place?

Ans. The lungs themselves cannot contract, but their movement depends on the diaphragm and intercostal muscles. During inhalation, the diaphragm contracts and flattens downward while the external intercostal muscles contract, pulling the ribs upward and outward. These changes increase the thoracic cavity volume, lowering the air pressure inside the lungs, causing air to rush in. During exhalation, the diaphragm and intercostals relax, returning to their normal positions. This decreases thoracic volume and increases air pressure in the lungs, forcing air out. During forced breathing, internal intercostals and abdominal muscles assist by pushing air out more forcefully.

MARKING SCHEME (3 Marks)

- Role of diaphragm in breathing – 1
- Role of intercostal muscles in inhalation – 1
- Relaxation leading to exhalation/forced expiration – 1

Q.8 What is chloride shift?

Ans. The chloride shift, also called the Hamburger phenomenon, is a vital process in carbon dioxide transport.



In the tissues, carbon dioxide diffuses into red blood cells and is converted into **bicarbonate ions (HCO_3^-)** by the enzyme carbonic anhydrase. To prevent accumulation, bicarbonate diffuses out of the red blood cell into the plasma. To maintain **electrical neutrality**, an equal number of **chloride ions (Cl^-)** move into the red blood cell from the plasma. This ion exchange balances the charges and keeps red blood cells stable. In the lungs, the process reverses to allow release of carbon dioxide for exhalation.

MARKING SCHEME (3 Marks)

- Definition of chloride shift – 1
- Movement of bicarbonate out of RBC – 1
- Influx of chloride ions to maintain neutrality – 1

Q.9 What are the advantages of having millions of alveoli rather than a pair of simple balloon-like lungs?

Ans. The presence of millions of alveoli provides the lungs with an **enormous surface area** for gaseous exchange, far greater than what a pair of simple balloon-like lungs could provide. This ensures sufficient **oxygen uptake** to meet the high energy demands of human cells and efficient **carbon dioxide elimination**. The small size of alveoli means that each is in **close contact with a dense capillary network**, reducing the distance for diffusion and speeding up gas exchange. Moreover, having many small alveoli allows the lungs to remain **elastic and flexible**, accommodating large volumes of air without damage. This design ensures **maximum respiratory efficiency**.

MARKING SCHEME (3 Marks)

- Increased surface area for gas exchange – 1
- Efficient diffusion due to thin walls and capillary contact – 1
- Meeting high metabolic/respiratory demands – 1

Q.10 Write the differentiate between:

(a) Internal and External Respiration.

Ans. **External respiration** occurs in the lungs and involves the exchange of **gases between alveolar air and blood**. Oxygen diffuses from alveoli into blood, while carbon dioxide diffuses from blood into alveoli for exhalation.

Internal respiration takes place in tissues, where oxygen carried by haemoglobin is released into body cells, and carbon dioxide produced by metabolism diffuses from cells into blood.

Thus, external respiration ensures oxygen loading in the lungs, while internal respiration ensures oxygen unloading in tissues. Both processes are vital for maintaining continuous cellular respiration and energy release.

MARKING SCHEME (3 Marks)

- Definition of external respiration – 1
- Definition of internal respiration – 1
- Difference in site/process clearly explained – 1

(b) Upper and Lower Respiratory Tract.

Ans. The **upper respiratory tract** includes the nose, nasal cavity, pharynx, and larynx. Its main functions are to filter, warm, and moisten incoming air and to act as a passageway to the lower tract.

The **lower respiratory tract** begins with the trachea, branching into **bronchi, bronchioles, and alveoli** inside the lungs. Its primary role is the **actual gas exchange between air and blood**.

In short, the upper tract prepares air for entry, while the lower tract provides the respiratory surface. Together, they maintain an efficient respiratory system.

MARKING SCHEME (3 Marks)

- Components of upper tract – 1
- Components of lower tract – 1
- Difference in function – 1

(c) Bronchi and Bronchioles.

Ans. The **bronchi** are two large tubes branching from the trachea into the right and left lungs. They have **cartilage rings** in their walls to maintain shape and prevent collapse. Their function is to conduct air deeper into the lungs.

The **bronchioles** are finer branches of the bronchi that lack cartilage and instead have smooth muscle. Their narrow structure allows them to regulate airflow by constriction or dilation. They terminate in clusters of **alveoli**, where gas exchange occurs.

Thus, bronchi act as main air passageways, while bronchioles serve as **regulators of airflow and connect to the alveoli**.

MARKING SCHEME (3 Marks)

- Structure of bronchi (large tubes with cartilage) – 1
- Structure of bronchioles (small tubes, smooth muscle) – 1
- Functional difference (air passage vs regulation and alveoli connection) – 1

(d) Oxyhaemoglobin and Carboxyhaemoglobin.

Ans. **Oxyhaemoglobin** is formed when oxygen binds reversibly with haemoglobin inside red blood cells. It plays a vital role in transporting oxygen from the lungs to body tissues, where oxygen is released for cellular respiration. This binding is reversible and supports life.

MARKING SCHEME (3 Marks)

- Oxyhaemoglobin: definition and role – 1
- Carboxyhaemoglobin: definition and danger – 1
- Clear distinction (reversible vs irreversible binding) – 1



Carboxyhaemoglobin, on the other hand, is formed when carbon monoxide (CO) binds irreversibly with haemoglobin. CO has about **200 times greater affinity** for haemoglobin than oxygen, making it highly dangerous. This prevents haemoglobin from carrying oxygen, leading to tissue hypoxia, suffocation, and even death.

Thus, oxyhaemoglobin is essential for survival, while carboxyhaemoglobin is harmful.

(e) Haemoglobin and Myoglobin.

Ans. Haemoglobin is a red, iron-containing protein found in red blood cells. It has **four polypeptide chains** and can bind up to **four molecules of oxygen**. Its main function is the transport of oxygen from the lungs to tissues and carbon dioxide back to the lungs.

Myoglobin is a similar oxygen-binding protein but found in **muscle cells**.

It consists of a **single polypeptide chain** and can bind only one oxygen molecule.

Its function is to **store oxygen** within muscle fibers and release it during intense muscular activity. Therefore, haemoglobin transports oxygen in blood, while myoglobin stores oxygen in muscles for immediate use.

MARKING SCHEME (3 Marks)

- Structure and function of haemoglobin - 1
- Structure and function of myoglobin - 1
- Key difference (oxygen transport vs storage) - 1

Q.11 List the effects of smoking.

Ans. Smoking has serious harmful effects on the respiratory system and general health. It is responsible for about **87% of lung cancer cases** and also increases the risk of cancers of the **mouth, larynx, and esophagus**.

It is a major cause of **chronic bronchitis and emphysema**, two diseases that permanently damage the lungs. Smoking irritates the airways, producing a persistent **morning cough**, and damages the cilia that normally help remove dust and microbes, leaving the body vulnerable to infections like **pneumonia**.

Almost immediately, smoking can make breathing difficult and, over time, it worsens **asthma and allergies**. Thus, smoking severely impairs respiratory efficiency and reduces life expectancy.

MARKING SCHEME (3 Marks)

- Cancer-related effects - 1
- Chronic respiratory diseases (bronchitis, emphysema, asthma, allergies) - 1
- Cough, cilia damage, infections / difficulty in breathing - 1

III

Long Answer Questions

Q.1 Describe the human upper respiratory tract.

Ans. Please See SLO [B-12-R-02]

Q.2 Describe the human lower respiratory tract.

Ans. Please See SLO [B-12-R-02]

Q.3 Describe the mechanism of breathing in man.

Ans. Please See SLO [B-12-R-03]

Q.4 How the control of breathing takes place?

Ans. Please See SLO [B-12-R-03]

Q.5 Explain the transport of oxygen in blood.

Ans. Please See SLO [B-12-R-04]

Q.6 Explain the transport of carbon dioxide in blood.

Ans. Please See SLO [B-12-R-04]

Q.7 What is the role of respiratory pigments in man?

Ans. Please See SLO [B-12-R-05]

Q.8 Describe the cause, symptoms and treatments of:

(a) Sinusitis (b) Otitis media (c) Pneumonia (d) Pulmonary tuberculosis

(e) Emphysema (f) Chronic obstructive pulmonary disease (COPD)

Ans. Please See SLO [B-12-R-06]

